

A CHARACTERIZATION OF VARIOUS SEIDEL EQUIVALENCE CLASSES OF GRAPHS

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Abstract Let $G = (V, E)$ be a graph with adjacency matrix $A(G)$ and Seidel matrix $S(G) = J - 2A(G) - I$. Influenced by results for adjacency matrices and the adjacency spectrum, we examine the characteristic polynomial $\phi(S(G))$ of the Seidel matrix. We provide relevant mathematical background then proceed to introduce original results. We prove a general form for the coefficients of some terms of $\phi(S(G))$ for all graphs on n vertices. We also prove a general form for the Seidel characteristic polynomial for the complete bipartite switching class and its complement.

Introduction

In this project we approach studying graph theory through a linear algebraic lens. Graph theory is a sub-field of combinatorics in which we can study the relationships between different structures. The graphical representation of data structures allows for their detailed mathematical analysis which is useful in many applications of computer science, including the construction of neural networks and the study of transit systems. One particularly useful way to represent the structural arrangement of a graph is by encoding this data as a matrix. Once in matrix form, the field of linear algebra can be applied to better study specific graph properties.

In the first five sections of this paper we give relevant background information and results on topics such as graph theory, combinatorics, linear algebra, and abstract algebra. Additionally, we give detailed proofs of some well established results in linear algebra and spectral graph theory.

In Section 6 we discuss original research regarding the Seidel characteristic polynomial and spectrum of graphs. We then examine the organization of graphs through their relationship to the Seidel matrix. We conclude this paper with proposed conjectures and directions for future research.

1 Graph preliminaries

A *graph* is a tuple $G = (V(G), E(G))$ where $V(G)$ denotes a set of vertices of G and $E(G)$ denotes a set of 2-element subsets of $V(G)$, indicating edges of G . For $v_1, v_2 \in V(G)$, v_1 and v_2 are *adjacent* if $\{v_1, v_2\} \in E(G)$. A *simple* graph is a graph containing no loops or multiple edges between the same pair of vertices; all graphs referred to here will be simple, undirected graphs. Thus any edge $\{v_i, v_j\} \in E(G)$ can be equivalently expressed as $\{v_j, v_i\}$ [4]. We will use n to denote the number of vertices in a graph G . The *degree* of a vertex v is the number of edges incident to v .

Consider the graph G in Figure 1 with $n = 4$ vertices where vertices of the same color have the same degree. We see that vertex d has degree 1, a has degree 3, and b and c both have degree 2. When all n vertices in G have the same degree k , we say that G is a *k -regular graph*.

Consider graphs G and H in Figure 1. Since G has vertices of different degrees it is not a regular graph. H has all vertices with degree 2 so H is 2-regular.

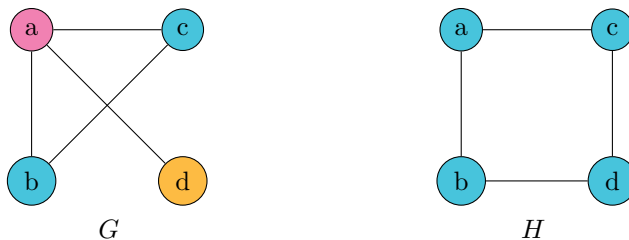


Figure 1: non-regular graph (G) and 2-regular graph (H)

Next we will define terms relating to graph traversal and see a small example. A *walk* on a graph is an

alternating list of vertices and edges, a *trail* is a walk with no repeated edges, and a *path* is a trail with no repeated vertices. The *distance*, $\text{dist}(v_i, v_j)$ of a pair of vertices is the length of the shortest path between them. For the graph G in Figure 1 the distance between vertices b and d is 2. The *diameter* is the maximum distance between any pair of vertices of G . A graph is *connected* if there exists a walk between any two vertices. Otherwise, we say that a graph is *disconnected*.

The remainder of this section will provide specifications for well known types of graphs and their properties. See Figure 2 for examples of each type of graph. A graph with an empty edge set is called an *empty graph* E_n . In a *complete graph* K_n , every pair of vertices has an edge between them. A *cycle graph* C_n is a connected graph such that all vertices have degree 2. A *path graph* P_n contains $n-2$ vertices of degree 2 and two vertices of degree 1. We can also define a path by its construction: it is a graph formed by removing one edge from C_n .

A graph whose vertex set can be partitioned into subsets A and B with respective orders a and b such that no two vertices of the same set are adjacent is known as a *bipartite graph*. In a *complete bipartite graph* $K_{a,b}$, every vertex in A is adjacent to every vertex in B . A subclass of bipartite graphs are trees. A *tree* is a connected graph containing no cycles. Any vertex of degree 1 in a tree is called a *leaf*. A graph made up of a disjoint union of trees is called a *forest*. A path graph is an example of a tree and has two leaves.

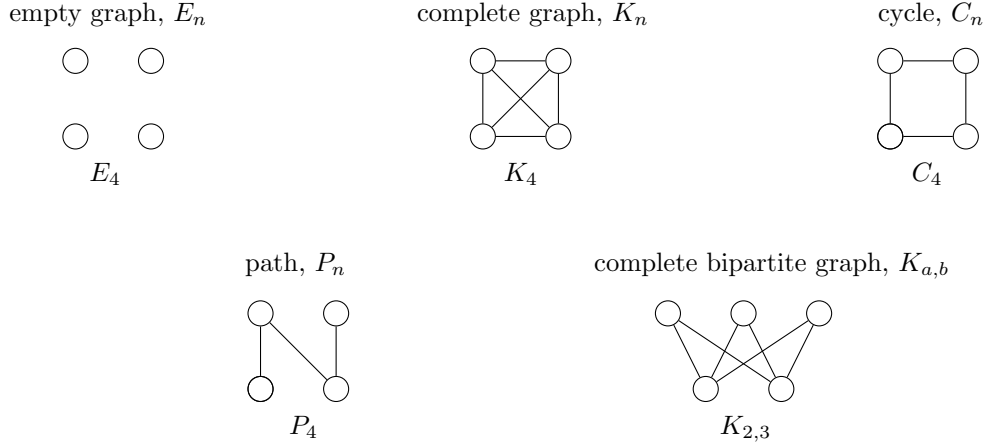


Figure 2: common graph examples

1.1 Graph complements

One structural process we can examine is taking the complement of a graph. Consider a graph G with n vertices and edge set $E(G)$. The *graph complement* of G , denoted $\bar{G}$ is the graph with edge set $E(\bar{G})$. $\bar{G}$ is defined such that $V(G) = V(\bar{G})$ and $E(G) \cup E(\bar{G}) = E(K_n)$ and $E(G) \cap E(\bar{G}) = \emptyset$ where $E(K_n)$ gives all possible edges for a graph on n vertices. We can observe that superimposing G and $\bar{G}$ gives the complete graph K_n . For a trivial example, consider the superimposition of graph complements E_n and K_n . Next consider the superimposition of graph complements $K_2 + K_2$ and C_4 as shown in Figure 3.

There exist methods for finding the number of edges present in K_n other than superimposing graph complements. We can observe from Figure 3 that $|E(G)| + |E(\bar{G})| = |E(K_n)| = \binom{n}{2}$, since the number of edges in a complete graph is equal to the counting number $\binom{n}{2}$ which is called the *binomial coefficient* $\binom{n}{k}$ and is given by $\frac{n!}{k!(n-k)!}$ for integers $0 \leq k \leq n$.

1.2 Subgraphs

Another tool for the examination of graphs is to find and study their subgraphs. Consider graphs G and H . If $V(H) \subseteq V(G)$ and $E(H) \subseteq E(G)$ then we call H a *subgraph* of G . A subgraph H of graph G is an *induced*

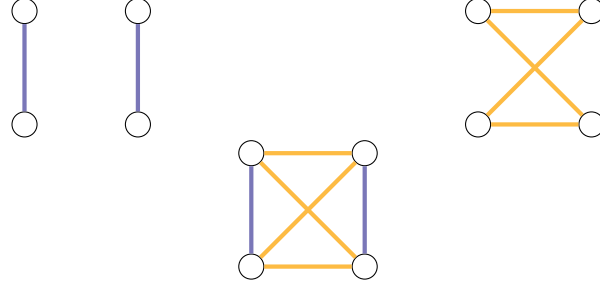


Figure 3: The superimposition of $K_2 + K_2$ and C_4 is K_4

subgraph if $E(H)$ contains all of $E(G)$ which have both endpoints in $V(H)$. A *component* of a graph G is a maximal connected subgraph of G .

An example of a graph with multiple components is given in Figure 4. We use a $(+)$ to denote multiple components in a graph. Let $G = K_2 + K_{1,3}$. Then $K_{1,3}$ is a component of G . The subgraph P_3 is not a component of G because it is not a maximal connected subgraph.

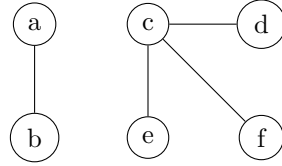


Figure 4: G made up of components K_2 with vertex set $\{a, b\}$ and $K_{1,3}$ with vertex set $\{c, d, e, f\}$

1.3 Graph isomorphisms

An important skill while studying graphs is the ability to detect distinct graphs from those which are structurally identical in order to better classify them. We say that given two graphs G and G' , if there exists a mapping of G' that preserves the adjacencies of G then we say that G' is an *isomorphism* of G , or that the graphs G and G' are *isomorphic*. We are interested in examining uniquely structured graphs; therefore, it is important that we are able to distinguish isomorphisms from uniquely structured graphs in order to exclude them from our data set.

Consider the graphs in Figure 5. When we rearrange vertices a, b, c, d of both graphs so that they are in the same positions, we see that we have an identical graph structure.



Figure 5: C_4 isomorphisms

2 Linear algebra preliminaries

This project examines topics in the intersection of graph theory and linear algebra. In this section we provide definitions for relevant matrix types and their properties adapted from *Linear Algebra and applications: An inquiry-based approach* by Alayont and Schlicker [1] and *Matrix analysis and applied linear algebra* by Meyer [7]. We then perform calculations to find information which characterizes a matrix which will have implications for graph structure in future sections. We conclude this section with results on matrix symmetry and invertibility.

Consider an $n \times n$ matrix $A = [a]_{ij}$ where i and j denote row and column positions, respectively. The *main diagonal* of A consists of entries $[a]_{ii}$ where the row and column numbers are equal ($j = i$). The *trace* of a matrix is the sum of its main diagonal entries, denoted

$$\text{tr}(A) = \sum_{i=1}^n [a]_{ii}.$$

We will use I_n to denote the identity matrix and J_n to denote the square matrix where every entry has value 1. When the dimensions are clear from context, we simply use I or J .

A matrix A is a *diagonal matrix* if $a_{ij} = 0$ for all $i \neq j$ and is *upper or lower triangular* if all entries of A are zero below and above the main diagonal respectively. The *transpose* of an $m \times n$ matrix $A = [a_{ij}]$ is the $n \times m$ matrix A^T whose i, j th entry is a_{ji} [1]. A matrix A is *symmetric* if it is equal to its transpose, that is, $A = A^T$. When an $n \times n$ matrix A is invertible, a unique A^{-1} exists such that $AA^{-1} = A^{-1}A = I$, and we say that A is *nonsingular*. If A is not invertible, then A is *singular*. An orthogonal matrix P is an $n \times n$ matrix such that $P^T = P^{-1}$ and thus $P^T P = I$.

While we can often invert a matrix using row reduction and other common methods, another method from which we can approach the problem of inverting a matrix is to apply the formula in Lemma 2.1. This is particularly useful since computing inverses of matrices can be a lengthy process so by decomposing this operation into more manageable pieces we can simplify our computations.

Lemma 2.1. (Miller [8]) *Let Q be matrix and let P be an $n \times n$ nonsingular matrix. If P and $P + Q$ are invertible and Q has rank 1 then,*

$$(P + Q)^{-1} = P^{-1} - \frac{1}{1 + \text{tr}(QP^{-1})} P^{-1} Q P^{-1}.$$

2.1 The determinant

A property that can often be used to describe important information about square matrices is the determinant. If A is an $n \times n$ matrix, the *determinant* of A is the scalar $\det(A) = a_{11}C_{11} + a_{12}C_{12} + \dots + a_{1n}C_{1n}$ where $C_{ij} = (-1)^{i+j} \det(A_{ij})$ is the ij -cofactor of A , and A_{ij} is the matrix obtained by removing row i and column j of matrix A . If A is upper or lower triangular, then $\det(A)$ is the product of the main diagonal entries of A , or $\det(A) = \prod_{i=1}^n a_{ii}$.

In terms of row operations, adding a multiple of a row to another row does not change the determinant of a matrix. However, multiplying a row by a constant c multiplies the determinant by c , and swapping two rows multiplies the determinant by (-1) . The determinant can be used to describe specific matrix properties including invertibility. For example A is invertible if and only if $\det(A) \neq 0$.

Given $n \times n$ matrices A and B the following properties hold:

- (1) $\det(AB) = \det(A) \det(B)$
- (2) $\det(A^k) = (\det(A))^k$

$$(3) \det(A^T) = \det(A).$$

We can use the structure of block matrices in order to more efficiently perform operations such as finding the determinant. The following theorem provides a formula for the computation of the determinant of a block matrix.

Theorem 2.2. [7] Consider a matrix M with blocks A, B, C, D , where A and D are square matrices then

$$\det(M) = \det \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{cases} \det(A) \det(D - CA^{-1}B) & \text{when } A^{-1} \text{ exists,} \\ \det(D) \det(A - BD^{-1}C) & \text{when } D^{-1} \text{ exists.} \end{cases}$$

The Schur complements are the matrices $D - CA^{-1}B$ and $A - BD^{-1}C$ of A and D respectively.

2.2 Eigenvalues and the characteristic polynomial

We will make great use of the concept of eigenvalues which can represent various matrix properties. Therefore we will define relevant operations and follow an example to which we will apply our knowledge of the determinant. Consider a square matrix A and vector $\vec{v} \neq \vec{0}$. A scalar λ which satisfies the expression $A\vec{v} = \lambda\vec{v}$ is an *eigenvalue* of A and $\vec{v}$ is an *eigenvector* of A corresponding to λ . Any eigenvectors corresponding to distinct eigenvalues are linearly independent.

To calculate the eigenvalues of A we first calculate $\det(A - \lambda I)$. This expression is the *characteristic polynomial* of A , denoted $\phi(A)$. Then we can set the characteristic polynomial equal to 0 and solve for λ . Consider the matrix

$$A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

Then,

$$\begin{aligned} (A - \lambda I) &= \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} \\ &= \begin{bmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{bmatrix}. \end{aligned}$$

Thus,

$$\begin{aligned} \phi(A) &= \det(A - \lambda I) \\ &= \det \begin{bmatrix} -\lambda & 1 & 1 \\ 1 & -\lambda & 1 \\ 1 & 1 & -\lambda \end{bmatrix} \\ &= -\lambda(\lambda^2 - 1) - 1(-\lambda - 1) + 1(1 - (-\lambda)) \\ &= -\lambda(\lambda^2 - 1) + (\lambda + 1) + (1 + \lambda) \\ &= -\lambda^3 + \lambda + \lambda + 1 + 1 + \lambda \\ &= -\lambda^3 + 3\lambda + 2. \end{aligned}$$

When factored, this gives eigenvalues $\lambda_1 = 2, \lambda_2 = -1, \lambda_3 = -1$. Observe that $A_{3 \times 3}$ has 3 real eigenvalues. From here we can calculate the corresponding eigenvectors by finding solutions for $(A - \lambda I)\vec{v} = \vec{0}$. We proceed by row reducing the augmented matrix $A - 2I$,

$$\left[\begin{array}{ccc|c} -2 & 1 & 1 & 0 \\ 1 & -2 & 1 & 0 \\ 1 & 1 & -2 & 0 \end{array} \right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

which gives the eigenvector $\vec{1}_{3 \times 1}$. Next we find the eigenvectors corresponding with the eigenvalue -1 . We now row reduce the matrix $A + I$

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 \end{array} \right] \longrightarrow \left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

which gives the eigenvectors

$$\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \text{ and } \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}.$$

We can verify the eigenpairs of $(\lambda, \vec{v})$ by checking equality for $A\vec{v} = \lambda\vec{v}$.

Theorem 2.3. *An $n \times n$ matrix A that is real and symmetric has n real eigenvalues.*

Proof. Let A be a real matrix. Suppose λ with $\vec{v} \neq \vec{0}$ are an eigenvalue and eigenvector of A respectively with values in $\mathbb{C}$. Let $\vec{v}^*$ denote the conjugate transpose of $\vec{v}$.

Then $\vec{v}_{1 \times m} \vec{v}_{m \times 1}^* = \vec{r}_{1 \times 1}$ where $\vec{r}$ is real for any $\vec{v}$. By definition of eigenvalues, $\lambda\vec{v} = A\vec{v}$. So:

$$\begin{aligned} \vec{v}^*(\lambda\vec{v}) &= \vec{v}^*(A\vec{v}) \\ \lambda\vec{v}^*\vec{v} &= \vec{v}^*A\vec{v} \\ \lambda\vec{r} &= \vec{v}^*A\vec{v}. \end{aligned}$$

We proceed by taking the conjugate transpose of both sides $(\lambda\vec{r})^* = (\vec{v}^*A\vec{v})^*$. Since $\vec{r}$ is real $\vec{r}^* = \vec{r}^T$ which is also real. Hence:

$$\begin{aligned} \lambda^* \vec{r}^T &= (\vec{v}^*A\vec{v})^* \\ &= \vec{v}^* A^* \vec{v} \\ &= \vec{v}^* A^T \vec{v} \quad \text{since } A \text{ is real} \\ &= \vec{v}^* A \vec{v} \quad \text{since } A \text{ is symmetric} \\ &= \lambda \vec{r} \quad \text{by substitution.} \end{aligned}$$

Thus $\lambda^* = \lambda$. Since this property holds only for real numbers, eigenvalues λ of a real and symmetric matrix will be real numbers. $\square$

2.3 Similar matrices

When we are comparing matrices we can often learn useful structural information about a matrix by determining if it is similar to another matrix. We say that $n \times n$ matrices A and B are *similar* if there exists a nonsingular matrix P such that $B = P^{-1}AP$. Properties of similar matrices include equivalent characteristic polynomials, trace, determinants, and eigenvalues. That is, we can show that two matrices have the same multiset of eigenvalues by showing that they are similar.

3 Abstract algebra preliminaries

In this section we examine permutations and equivalence relations and use these structures to solve problems from linear algebra. Definitions are adapted from *Matrix analysis and applied linear algebra* by Meyer [7]. We conclude this section with an algebraic examination of the structural relationships between graphs.

3.1 Permutations

We now proceed to study permutations and their various properties, some of which lend well to the study of graphs. Permutations as seen through a more combinatorial lens can be considered to be an arrangement where order matters. In terms of abstract algebra, a *permutation* on $[n] = \{1, 2, \dots, n\}$ is a bijection $\sigma : [n] \rightarrow [n]$. The set of bijections from $[n]$ to $[n]$ forms the symmetric group S_n . Note that there exist $n!$ total permutations on $[n]$. We can write each of these orderings of elements in a form known as cycle notation which gives a map between elements. It is known that every permutation can be written as a composition of 2-cycles. Note that in σ_1 , the identity permutation, all elements map to themselves so it is a composition of zero 2-cycles. See Figure 6 for details of the group S_3 .

permutation	ordering	cycle notation	composition of 2- cycles
σ_1	$\{123\}$	$(1)(2)(3)$	
σ_2	$\{132\}$	$(1)(23)$	(23)
σ_3	$\{213\}$	$(12)(3)$	(12)
σ_4	$\{231\}$	(132)	$(12)(13)$
σ_5	$\{312\}$	(123)	$(13)(12)$
σ_6	$\{321\}$	$(13)(2)$	(13)

Figure 6: S_3 : the permutation group on the set $[3]$

Consider σ_3 . We read the cycle as indicating element 1 maps to element 2 which maps back to element 1, so we have a 2-cycle. As 3 maps to itself it is in a 1-cycle. Observe that 1-cycles are elements that are fixed by a permutation.

Every permutation has an associated property known as the sign of a permutation, denoted $\text{sgn}(\sigma)$ such that $\text{sgn}(\sigma) = (-1)^{n-k}$ where k is the number of disjoint cycles of any length in σ . Equivalently, $\text{sgn}(\sigma)$ is equal to negative one raised to the number of 2-cycles. If $\text{sgn}(\sigma) = -1$ then σ is an *odd permutation*, and if $\text{sgn}(\sigma) = 1$ then σ is an *even permutation*.

3.2 Permutation definition of the determinant

As an alternative to linear algebra techniques, we can use permutations and their properties to calculate the determinant of a matrix A with dimensions $n \times n$, following the formula:

$$\det(A) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) (a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{n,\sigma(n)}).$$

permutation	$\text{sgn}(\sigma)$	$(a_{1,\pi} \cdots a_{n,\pi})$	product	value
σ_1	1	$(a_{11}a_{22}a_{33})$	$1 \cdot (-\lambda)(-\lambda)(-\lambda)$	$-\lambda^3$
σ_2	-1	$(a_{11}a_{23}a_{32})$	$-1 \cdot (-\lambda)(1)(1)$	λ
σ_3	-1	$(a_{12}a_{21}a_{33})$	$-1 \cdot (1)(1)(-\lambda)$	λ
σ_4	1	$(a_{12}a_{23}a_{31})$	$1 \cdot (1)(1)(0)$	0
σ_5	1	$(a_{13}a_{21}a_{32})$	$1 \cdot (0)(1)(1)$	0
σ_6	-1	$(a_{13}a_{22}a_{31})$	$-1 \cdot (0)(-\lambda)(0)$	0

Figure 7: terms of $\phi(A)$

Consider using this permutation definition of the determinant to find the characteristic polynomial of the matrix

$$A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

To find $\phi(A)$ we find the determinant of the matrix:

$$A - \lambda I = \begin{bmatrix} -\lambda & 1 & 0 \\ 1 & -\lambda & 1 \\ 0 & 1 & -\lambda \end{bmatrix}.$$

To find the determinant of the 3×3 matrix A we must use S_3 . Let $i = 1, \dots, 6$ enumerate the permutations of S_3 . We proceed by finding the product of $\text{sgn}(\sigma_i)$ and the matrix entries and then sum these products to find the value of the determinant seen in Figure 7. The sum of the rightmost column gives the determinant of $A - \lambda I$. That is, we have found $\phi(A) = -\lambda^3 + 2\lambda$.

3.3 Equivalence relations

Other algebraic structures present in our results are equivalence relations on sets of graphs. Let $\mathcal{G}_n$ be the set of graphs on n vertices. Let G, G', G'' be graphs in $\mathcal{G}_n$. An equivalence relation $\sim$ on $\mathcal{G}_n$ is a relation from $\mathcal{G}_n$ to $\mathcal{G}_n$ that satisfies,

- i. $G \sim G$
- ii. if $G \sim G'$ then $G' \sim G$
- iii. if $G \sim G'$ and $G' \sim G''$ then $G \sim G''$

An equivalence relation will partition a set into *equivalence classes*. For an example, matrix similarity, as defined in Section 2.3, is an equivalence relation on the set of square matrices. Graph isomorphisms, as defined in Section 1.3, also form equivalence classes.

4 Spectral graph theory

In addition to properties defined in Section 1, we also study graph properties found through matrix operations. We apply some applications of linear algebra to the study of graph structure.

Given a graph G we can construct a matrix $A(G)$ encoding the structure of G . We have seen that we can calculate the eigenvalues of a matrix. Note that when we examine a graph matrix we refer to the eigenvalues of this matrix as the eigenvalues of the graph.

While a variety of graph matrices exist, in the following section we will focus on the adjacency matrix, which is a very commonly studied matrix. In later sections, we will use known results on this matrix as motivation to examine the less commonly studied Seidel matrix.

4.1 The adjacency matrix

The *adjacency matrix* of a graph G with vertex set $\{v_1, v_2, \dots, v_n\}$ is the $n \times n$ matrix denoted $A(G) = [a]_{ij}$. The entries of the adjacency matrix are given by

$$a_{ij} = \begin{cases} 1 & \text{if } \{v_i, v_j\} \in E(G) \\ 0 & \text{if } \{v_i, v_j\} \notin E(G). \end{cases}$$

All simple, undirected graphs have symmetric adjacency matrices with $\text{tr}(A) = 0$.

The set containing the eigenvalues of the adjacency matrix of G and their multiplicities is known as the *adjacency spectrum* of G . Since the adjacency matrix A is real and symmetric, A has n real eigenvalues by Theorem 2.3. Consider the characteristic polynomial of K_3 given by $\phi(K_3) = \det(A(K_3) - \lambda I)$. The characteristic polynomial is calculated in Section 2.2 and gives $\phi(K_3) = -\lambda^3 + 3\lambda + 2$ which we factor to find eigenvalues $\lambda_1 = 2, \lambda_2 = -1, \lambda_3 = -1$. Thus the spectrum of K_3 is $\{2^{(1)}, -1^{(2)}\}$, where multiplicities are denoted by superscripts [6].

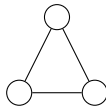


Figure 8: the graph K_3

There are a variety of structural graph properties that we understand from the adjacency spectrum and characteristic polynomial. The number of vertices of a graph G can be determined from the spectrum as it is equal to the number of eigenvalues present, which is equal to the degree of the adjacency characteristic polynomial. The number of edges in G is given by the magnitude of third coefficient of the characteristic polynomial. Consider the example in Figure 8. We can count 3 edges in our graph and observe that the coefficient of the degree $n - 2$ term in $\phi(K_3)$ has a magnitude of 3, matching the edge number.

4.1.1 Cospectral graphs

While certain structural properties of graphs can be determined from a graph's spectrum, there are other properties which cannot be determined in this way. Next we will examine graphs whose structural properties provide insight into what can be discerned about a graph by spectrum alone.

Graphs that share the same spectrum are said to be *cospectral*. All pairs of isomorphic graphs are cospectral to one another. Here, we will refer to non-isomorphic sets of graphs which share the same spectrum as cospectral graphs. Note that all graphs cospectral to one another with respect to the adjacency matrix must contain the same number of vertices and edges.

The smallest set of adjacency cospectral graphs has 5 vertices and has spectrum $\{2^{(1)}, 0^{(3)}, -2^{(1)}\}$. The two nonisomorphic graphs with this adjacency spectrum are given in Figure 9. This example illustrates that structural properties such as whether a graph is a tree or is connected cannot be implied from an examination of the spectrum alone.

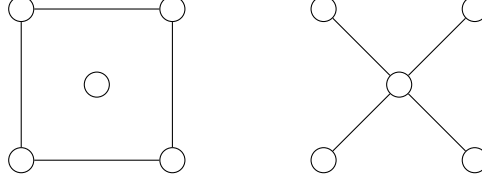


Figure 9: Saltire Pair

4.1.2 Graphs determined by their spectrum

We have seen that not all graph properties can be determined by a graph's spectrum. In this section we will investigate which graphs can be fully constructed from information in the spectrum. A graph G is said to be *determined by its spectrum (DS)* with respect to a matrix A or S if there does not exist any graph G' cospectral and nonisomorphic to G . The spectrum of a *DS* graph is unique to that graph, so if we know the spectrum, we can build only one uniquely structured graph. Equivalently all graphs with this spectrum are isomorphic to the original. The number of non-*DS* graphs with respect to A increases over $n = 5, \dots, 10$ vertices. However, the percentage of non-*DS* graphs on greater than 10 vertices appears to decrease according to computational evidence [3]. In general, it is easier to prove that graphs are not *DS* than it is to show that a graph is *DS*.

<i>DS</i> graph	spectrum
K_n	$\{(n-1)^{(1)}, (-1)^{(n-1)}\}$
E_n	$\{0^{(n)}\}$
C_n	$\{2 \cos(\frac{2k\pi}{n}) : k = 0, \dots, n-1\}$
P_n	$\{2 \cos(\frac{k\pi}{n+1}) : k = 1, \dots, n\}$

Figure 10: some common *DS* graphs

Note that the disjoint union of *DS* graphs does not necessarily create another *DS* graph. There are some exceptions, including the disjoint union of r path graphs $P_{n_1} + \dots + P_{n_r}$ which is *DS* with respect to A . This is also true for the disjoint union of r complete graphs $K_{n_1} + \dots + K_{n_r}$, and the disjoint union of cycle graphs $C_{n_1} + \dots + C_{n_r}$. Examples of *DS* graphs and their spectra are given in Figure 10.

Since there exist no adjacency cospectral pairs on fewer than 5 vertices, all graphs with 4 or fewer vertices are *DS*. In contrast, when we examine the Seidel matrix we find that for $n > 1$ vertices there are no Seidel-*DS*

graphs.

4.2 The Seidel matrix

The *Seidel matrix* of a graph G with vertex set $\{v_1, v_2, \dots, v_n\}$ is the $n \times n$ matrix denoted $S(G) = [s]_{ij}$. The entries of the Seidel matrix are given by:

$$s_{ij} = \begin{cases} -1 & \text{if } \{v_i, v_j\} \in E(G) \\ 1 & \text{if } \{v_i, v_j\} \notin E(G) \\ 0 & \text{if } v_i = v_j. \end{cases}$$

Equivalently the Seidel matrix S is given by $S = J - 2A - I$.

5 Seidel switching

In this section we define and compare specific types of switching operations on graphs and matrices. We conclude the section with a detailed proof of a well known switching result from spectral graph theory.

The Seidel matrix is strongly related to a structural operation on graphs known as Seidel switching which removes certain edges of a graph and then adds others. *Seidel switching* on a graph G constructs graphs which are cospectral to G with respect to the Seidel matrix. Seidel switching is an equivalence relation that determines equivalence classes, called *Seidel classes* or *switching classes*, for the set of graphs on n vertices. Switching equivalent graphs have similar Seidel matrices and therefore the same Seidel spectrum and characteristic polynomial. The switching classes for $\mathcal{G}_4$ are given in Figure 11.

All Seidel cospectral graphs possess the same number of vertices, but they may contain a different number of edges. Seidel switching which is described below can be accomplished through both matrix and graph switching to build each class of Seidel cospectral graphs.

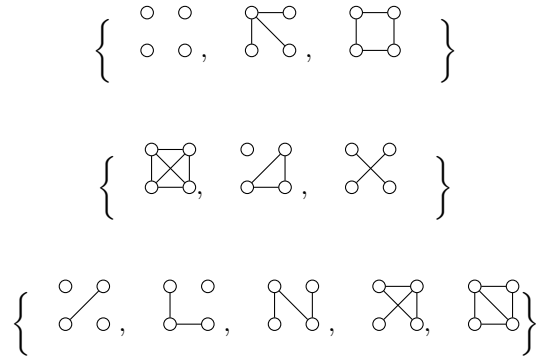


Figure 11: $\mathcal{G}_4$ Seidel classes

5.1 Matrix switching

There exist multiple methods for conducting a Seidel switch. We will define the process of matrix switching before characterizing another method. A *Seidel switch* requires the vertices of G to be partitioned into two vertex sets. We can then partition the Seidel matrix into submatrices corresponding to these vertex sets such that blocks S_{11} and S_{22} represent the adjacencies between vertices of the same vertex set and blocks S_{12} and S_{21} represent the adjacencies between vertices of different vertex sets, resulting in:

$$S(G) = \begin{bmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{bmatrix}.$$

Note that $S_{21} = S_{12}^T$ for all possible partitions; so we can rewrite $S(G)$ as follows:

$$S(G) = \begin{bmatrix} S_{11} & S_{12} \\ S_{12}^T & S_{22} \end{bmatrix}.$$

To perform a Seidel switch we replace blocks S_{12} and S_{12}^T with their negatives. This produces the Seidel matrix:

$$\tilde{S}(G) = \begin{bmatrix} S_{11} & -S_{12} \\ -S_{12}^T & S_{22} \end{bmatrix}.$$

Once a Seidel switch has been performed, we can find the graph that is switching equivalent to G , denoted $\tilde{G}$, using $\tilde{S}(G)$ to determine vertex adjacencies.

A graph $\tilde{G}$ that is cospectral to G with respect to the Seidel matrix is rarely an isomorphism of G . Since entries of -1 represent an edge between 2 vertices, while entries of 1 represent no edge between vertices, when the number of ± 1 entries varies from S to $\tilde{S}$ in the S_{12} block, the result is a that $\tilde{G}$ will possess a different number of edges than G , meaning no isomorphism is possible. $\tilde{G}$ will only possess the same number of edges as G in a special case which allows for the possibility of an isomorphism, when S_{12} has an equal number of 1 's and -1 's in the Seidel matrix $S(G)$.

There exist conditions on the graphs and on their vertex partitions where Seidel switching can also produce cospectral graphs with respect to the adjacency matrix, but in general this is not the case. We will discuss this special occurrence in Section 5.3.

For an example of Seidel switching using a matrix, we consider the graph $K_2 + K_2$. We proceed to perform a Seidel switch on the vertex set $\{a, b\}$, as defined in Figure 12, to find the graph K_4 to be Seidel cospectral to $K_2 + K_2$.

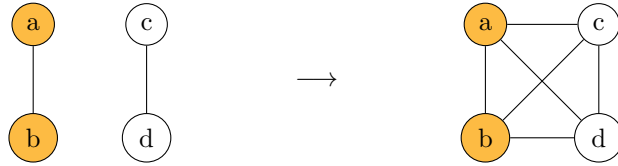


Figure 12: Seidel switch with switching set $\{a, b\}$

$$S(K_2 + K_2) = \left[\begin{array}{cc|cc} 0 & -1 & 1 & 1 \\ -1 & 0 & 1 & 1 \\ \hline 1 & 1 & 0 & -1 \\ 1 & 1 & -1 & 0 \end{array} \right]$$

$$= \left[\begin{array}{c|c} S_{11} & S_{12} \\ \hline S_{21}^T & S_{22} \end{array} \right].$$

Performing a Seidel switch, we have

$$\tilde{S} = \left[\begin{array}{c|c} S_{11} & -S_{12} \\ \hline -S_{21}^T & S_{22} \end{array} \right]$$

$$= \left[\begin{array}{cc|cc} 0 & -1 & -1 & -1 \\ -1 & 0 & -1 & -1 \\ \hline -1 & -1 & 0 & -1 \\ -1 & -1 & -1 & 0 \end{array} \right].$$

From $\tilde{S}$ we construct the graph K_4 as a Seidel cospectral graph to $K_2 + K_2$ with switching set $\{a, b\}$ as shown in Figure 12.

5.2 Graph switching

Similarly to matrix switching, *graph switching* allows us to construct a Seidel switching class, this time without the intermediate step of building matrices. Two graphs G and G_U are *switching equivalent* (or *Seidel equivalent*) if and only if G can be redrawn as G_U through Seidel switching operations.

Consider the partition of the vertices of G into sets U and $V - U$. Let G_U denote a graph obtained from G by switching with respect to U . Viewing a Seidel switch directly on the graph, we switch the adjacencies between all pairs (u, v) such that $u \in U$ and $v \in V - U$. This produces a switching equivalent graph, G_U that is equivalent to the the matrix switching process.

We will further detail the construction of a specific Seidel equivalence class composed of graphs in $\mathcal{G}_4$ in Section 5.4. We define $\mathcal{G}_4$ to be the set of uniquely structured graphs on 4 vertices in Figure 11. Note that the Seidel equivalence classes on $\mathcal{G}_4$ can be found by switching using either a matrix or a graph. We can relate graph switching to matrix switching when we observe that $\tilde{G}$ is equivalent to G_U where U contains all vertices that make up the S_{11} block of matrix $S(G)$ given above. So we see that the organization of the Seidel matrix lends well to graph structure.

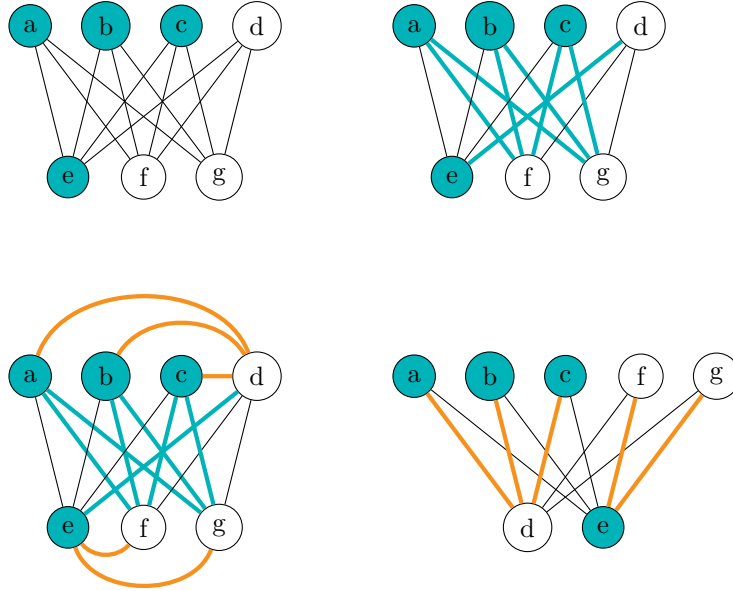


Figure 13: Seidel switch of $K_{3,4}$ to $K_{2,5}$ with switching set $\{a, b, c, e\}$, teal edges removed, and orange edges added

For an example of the switching process directly on a graph, consider performing a Seidel switch on the complete bipartite graph $K_{3,4}$. We use switching set $\{a, b, c, e\}$ highlighted in Figure 13. We proceed to remove all edges across vertex sets $\{a, b, c, e\}$ and $\{d, f, g\}$. Then we add any edges that were not previously present across our switching sets. When we reorder our vertices, we see that we have switched to the complete

bipartite graph $K_{2,5}$. We will prove in Section 6 that a Seidel switch performed on a complete bipartite graph will always generate another complete bipartite graph.

5.3 Godsil-McKay switching

There are many constructions of cospectral pairs that are currently known, including Seidel switching and Godsil-McKay switching. A *Godsil-McKay switch* (GM switch) is a type of Seidel switch used to construct adjacency cospectral graphs G and G' beginning with a graph G that satisfies certain conditions. This construction can be thought of as a form of Seidel Switching such that the graph produced is always cospectral with respect to both the adjacency and Seidel matrices.

Note that the smallest adjacency cospectral graphs that can be constructed by a GM-switch graphs are on 8 vertices, indicating that not every adjacency cospectral pair is obtainable through a GM switch.

To construct cospectral graphs using a GM switch we start with a graph G and let π be a partition of the vertex set of $V(G)$ such that $\pi = \{C_1, \dots, C_k, D\}$. Let i, j , and k be integers such that $1 \leq i$ and $j \leq k$. Since we will only consider this construction for the $k = 1$ case, we say that $i = 1 = k$. Then $\pi = (C, D)$. Let n denote the number of vertices within C . That is, $n = |C|$.

The specifications for performing a GM Switch are that a graph G must meet the following conditions:

- (a) n must be even.
- (b) The induced subgraph formed by the vertex set C must be a regular graph.
- (c) For all $v \in D$, v must be adjacent to either 0, $n/2$, or n vertices in C .

If conditions (a),(b), and (c) hold then for every $v \in D$ that has $n/2$ neighbors in C , we delete these edges and then add edges between v and the $n/2$ vertices in C to which v was not previously adjacent. After performing this process, we will have constructed a graph, denoted G^π , that is guaranteed to be adjacency cospectral to our original graph G . To detail the original proof given by Godsil and McKay in 1982 [5], we define a matrix Q_m then first show a series of lemmas.

Let $Q_m = (\frac{2}{m})J_m - I_m$ for any positive integer m . Then Q_m has the form:

$$\begin{aligned}
Q_m &= \left(\frac{2}{m}\right) J_m - I_m \\
&= \frac{2}{m} \begin{bmatrix} 1 & \dots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \dots & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & 1 \end{bmatrix} \\
&= \begin{bmatrix} (\frac{2}{m} - 1) & \frac{2}{m} & \dots & \frac{2}{m} \\ \frac{2}{m} & (\frac{2}{m} - 1) & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{2}{m} \\ \frac{2}{m} & \dots & \frac{2}{m} & (\frac{2}{m} - 1) \end{bmatrix}.
\end{aligned}$$

Lemma 5.1. $Q_m^2 = I_m$.

Proof.

$$\begin{aligned}
Q_m^2 &= \begin{bmatrix} \left(\frac{2}{m} - 1\right) & \frac{2}{m} & \cdots & \frac{2}{m} \\ \frac{2}{m} & \left(\frac{2}{m} - 1\right) & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{2}{m} \\ \frac{2}{m} & \cdots & \frac{2}{m} & \left(\frac{2}{m} - 1\right) \end{bmatrix}^2 \\
&= \begin{bmatrix} \left(\frac{2}{m} - 1\right) & \frac{2}{m} & \cdots & \frac{2}{m} \\ \frac{2}{m} & \left(\frac{2}{m} - 1\right) & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{2}{m} \\ \frac{2}{m} & \cdots & \frac{2}{m} & \left(\frac{2}{m} - 1\right) \end{bmatrix} \begin{bmatrix} \left(\frac{2}{m} - 1\right) & \frac{2}{m} & \cdots & \frac{2}{m} \\ \frac{2}{m} & \left(\frac{2}{m} - 1\right) & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{2}{m} \\ \frac{2}{m} & \cdots & \frac{2}{m} & \left(\frac{2}{m} - 1\right) \end{bmatrix}.
\end{aligned}$$

The entries on the main diagonal, $[Q_m^2]_{ii}$ will be in the form:

$$\left(\frac{2}{m} - 1\right) \left(\frac{2}{m} - 1\right) + \left(\frac{2}{m}\right) \left(\frac{2}{m}\right) + \dots + \left(\frac{2}{m}\right) \left(\frac{2}{m}\right)$$

where $\left(\frac{2}{m} - 1\right) \left(\frac{2}{m} - 1\right)$ is the i^{th} term of the sum. Then:

$$\begin{aligned}
\left(\frac{2}{m} - 1\right) \left(\frac{2}{m} - 1\right) + \left(\frac{2}{m}\right) \left(\frac{2}{m}\right) + \dots + \left(\frac{2}{m}\right) \left(\frac{2}{m}\right) &= \left(\frac{2}{m} - 1\right) \left(\frac{2}{m} - 1\right) + (m-1) \left(\frac{2}{m}\right) \left(\frac{2}{m}\right) \\
&= \left(\frac{2}{m} - 1\right)^2 + (m-1) \left(\frac{2}{m}\right)^2 \\
&= \left(\frac{4}{m^2} - \frac{2}{m} - \frac{2}{m} + 1\right) + (m-1) \left(\frac{4}{m^2}\right) \\
&= \frac{4}{m^2} - \frac{4}{m} + 1 + m \left(\frac{4}{m^2}\right) - \frac{4}{m^2} \\
&= -\frac{4}{m} + 1 + \frac{4}{m} \\
&= 1.
\end{aligned}$$

The off diagonal entries $[Q_m^2]_{ij}$ for $i \neq j$ have the form:

$$\begin{aligned}
\left(\frac{2}{m}\right) \left(\frac{2}{m} - 1\right) + \left(\frac{2}{m}\right) \left(\frac{2}{m} - 1\right) + \left(\frac{2}{m}\right) \left(\frac{2}{m}\right) + \dots + \left(\frac{2}{m}\right) \left(\frac{2}{m}\right) &= 2 \left(\frac{2}{m}\right) \left(\frac{2}{m} - 1\right) + (m-2) \left(\frac{2}{m}\right)^2 \\
&= 2 \left(\frac{4}{m^2} - \frac{2}{m}\right) + (m-2) \left(\frac{4}{m^2}\right)
\end{aligned}$$

$$= \frac{8}{m^2} - \frac{4}{m} + m \left(\frac{4}{m^2} \right) - \frac{8}{m^2}$$

$$= -\frac{4}{m} + \frac{4}{m}$$

$$= 0.$$

Then:

$$Q_m^2 = \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & 1 \end{bmatrix}$$

$$= I_m.$$

□

Lemma 5.2. *Let X be an $m \times n$ matrix with constant row sums r and constant column sums c . Then $Q_m X Q_n = X$.*

Proof. Without loss of generality, addition within the dot product can be reordered to show:

$$Q_m X Q_n = \begin{bmatrix} (\frac{2}{m} - 1) & \frac{2}{m} & \dots & \frac{2}{m} \\ \frac{2}{m} & (\frac{2}{m} - 1) & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{2}{m} \\ \frac{2}{m} & \dots & \frac{2}{m} & (\frac{2}{m} - 1) \end{bmatrix} \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} \begin{bmatrix} (\frac{2}{n} - 1) & \frac{2}{n} & \dots & \frac{2}{n} \\ \frac{2}{n} & (\frac{2}{n} - 1) & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{2}{n} \\ \frac{2}{n} & \dots & \frac{2}{n} & (\frac{2}{n} - 1) \end{bmatrix}$$

$$= \begin{bmatrix} a_{11}(\frac{2}{m} - 1) + a_{21}(\frac{2}{m}) + \dots + a_{m1}(\frac{2}{m}) & \dots & a_{1n}(\frac{2}{m} - 1) + a_{2n}(\frac{2}{m}) + \dots + a_{mn}(\frac{2}{m}) \\ \vdots & \ddots & \vdots \\ a_{m1}(\frac{2}{m} - 1) + a_{11}(\frac{2}{m}) + \dots + a_{(m-1)1}(\frac{2}{m}) & \dots & a_{mn}(\frac{2}{m} - 1) + a_{1n}(\frac{2}{m}) + \dots + a_{(m-1)n}(\frac{2}{m}) \end{bmatrix} Q_n$$

$$= \begin{bmatrix} a_{11}(\frac{2}{m} - 1) + (\frac{2}{m})(a_{21} + \dots + a_{m1}) & \dots & a_{1n}(\frac{2}{m} - 1) + (\frac{2}{m})(a_{2n} + \dots + a_{mn}) \\ \vdots & \ddots & \vdots \\ a_{m1}(\frac{2}{m} - 1) + (\frac{2}{m})(a_{11} + \dots + a_{(m-1)1}) & \dots & a_{mn}(\frac{2}{m} - 1) + (\frac{2}{m})(a_{1n} + \dots + a_{(m-1)n}) \end{bmatrix} Q_n.$$

Since the $m \times n$ matrix X has constant column sums c , without loss of generality, we can simplify the terms in the matrix $Q_m X$ of the form where:

$$[Q_m X]_{11} = a_{11} \left(\frac{2}{m} - 1 \right) + a_{21} \left(\frac{2}{m} \right) + \dots + a_{m1} \left(\frac{2}{m} \right)$$

$$\begin{aligned}
&= a_{11} \left(\frac{2}{m} - 1 \right) + \left(\frac{2}{m} \right) (a_{21} + \dots + a_{m1}) \\
&= a_{11} \left(\frac{2}{m} - 1 \right) + \left(\frac{2}{m} \right) (c - a_{11}) \\
&= \frac{2a_{11}}{m} - a_{11} + \frac{2c}{m} - \frac{2a_{11}}{m} \\
&= -a_{11} + \frac{2c}{m}.
\end{aligned}$$

so $[Q_m X]_{ij} = \frac{2c}{m} - a_{ij}$. Then:

$$Q_m X Q_n = \begin{bmatrix} -a_{11} + \frac{2c}{m} & \dots & -a_{1n} + \frac{2c}{m} \\ \vdots & \ddots & \vdots \\ -a_{m1} + \frac{2c}{m} & \dots & -a_{mn} + \frac{2c}{m} \end{bmatrix} \begin{bmatrix} \left(\frac{2}{n} - 1 \right) & \frac{2}{n} & \dots & \frac{2}{n} \\ \frac{2}{n} & \left(\frac{2}{n} - 1 \right) & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{2}{n} \\ \frac{2}{n} & \dots & \frac{2}{n} & \left(\frac{2}{n} - 1 \right) \end{bmatrix}.$$

Since X has constant row sums, without loss of generality by symmetry, the entries of the matrix $(Q_m X)Q_n$ can be expressed in the form:

$$\begin{aligned}
[Q_m X Q_n]_{11} &= \left(\frac{2}{n} - 1 \right) \left(-a_{11} + \frac{2c}{m} \right) + \frac{2}{n} \left(-a_{12} + \frac{2c}{m} \right) + \dots + \frac{2}{n} \left(-a_{1n} + \frac{2c}{m} \right) \\
&= \left(\frac{2}{n} - 1 \right) \left(-a_{11} + \frac{2c}{m} \right) + \frac{2}{n} \left(\left(-a_{12} + \frac{2c}{m} \right) + \dots + \left(-a_{1n} + \frac{2c}{m} \right) \right) \\
&= \left(\frac{2}{n} - 1 \right) \left(-a_{11} + \frac{2c}{m} \right) + \frac{2}{n} \left((-a_{12} + \dots - a_{1n}) + \left(\frac{2c}{m} \right) (n-1) \right) \\
&= \left(\frac{2}{n} - 1 \right) \left(-a_{11} + \frac{2c}{m} \right) + \frac{2}{n} \left((-r + a_{11}) + \left(n \frac{2c}{m} - \frac{2c}{m} \right) \right) \\
&= \left(\frac{2}{n} - 1 \right) \left(-a_{11} + \frac{2c}{m} \right) + \frac{2}{n} (-r + a_{11}) + \frac{2}{n} \left(n \frac{2c}{m} - \frac{2c}{m} \right) \\
&= -\frac{2a_{11}}{n} + \frac{2}{n} \left(\frac{2c}{m} \right) + a_{11} - \frac{2c}{m} - \frac{2r}{n} + \frac{2a_{11}}{n} + \frac{2}{n} \left(n \frac{2c}{m} \right) - \frac{2}{n} \left(\frac{2c}{m} \right) \\
&= a_{11} - \frac{2c}{m} - \frac{2r}{n} + \frac{4c}{m}
\end{aligned}$$

$$= a_{11} + \frac{2c}{m} - \frac{2r}{n}.$$

Since the adjacency matrix is square and all submatrices corresponding to subgraphs are also square and symmetric, $m = n$. Additionally, an adjacency matrix X will have its constant column sums equal to its constant row sums so that $c = r$. Thus,

$$\begin{aligned} a_{11} + \frac{2c}{m} - \frac{2r}{n} &= a_{11} + \frac{2c}{m} - \frac{2c}{m} \\ &= a_{11}. \end{aligned}$$

So:

$$\begin{aligned} Q_m X Q_n &= \begin{bmatrix} a_{11} & \dots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{m1} & \dots & a_{mn} \end{bmatrix} \\ &= X. \end{aligned}$$

□

Lemma 5.3. *Let Q_{2m} be a matrix with dimensions $2m \times 2m$. Let $\vec{x}$ be a vector with $2m$ entries such that m entries are 0 and m entries are 1. Let $\vec{j}_{2m}$ denote the all ones vector of dimension $2m$. Then $Q_{2m}\vec{x} = \vec{j}_{2m} - \vec{x}$.*

Proof. Recall that matrix Q is defined such that

$$\begin{aligned} Q_{2m} &= \left(\frac{2}{2m} \right) J_{2m} - I_{2m} \\ &= \begin{bmatrix} \frac{1}{m} & \dots & \frac{1}{m} \\ \vdots & \ddots & \vdots \\ \frac{1}{m} & \dots & \frac{1}{m} \end{bmatrix} - \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \dots & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{m} - 1 & \frac{1}{m} & \dots & \frac{1}{m} \\ \frac{1}{m} & \frac{1}{m} - 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{1}{m} \\ \frac{1}{m} & \dots & \frac{1}{m} & \frac{1}{m} - 1 \end{bmatrix}. \end{aligned}$$

Since entries $0_1, \dots, 0_m$ and $1_1, \dots, 1_m$ can be written in any order within $\vec{x}$, without loss of generality let

$$\vec{x} = \begin{bmatrix} 0_1 \\ \vdots \\ 0_m \\ 1_1 \\ \vdots \\ 1_m \end{bmatrix}.$$

Then:

$$\begin{aligned}
Q_{2m}\vec{x} &= \begin{bmatrix} \frac{1}{m} - 1 & \frac{1}{m} & \cdots & \frac{1}{m} \\ \frac{1}{m} & \frac{1}{m} - 1 & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{1}{m} \\ \frac{1}{m} & \cdots & \frac{1}{m} & \frac{1}{m} - 1 \end{bmatrix} \begin{bmatrix} 0_1 \\ \vdots \\ 0_m \\ 1_1 \\ \vdots \\ 1_m \end{bmatrix} \\
&= \begin{bmatrix} 0_1 + \dots + 0_m + 1_1(\frac{1}{m}) + \dots + 1_m(\frac{1}{m}) \\ \vdots \\ 0_1 + \dots + 0_m + 1_1(\frac{1}{m} - 1) + 1_2(\frac{1}{m}) + \dots + 1_m(\frac{1}{m}) \\ \vdots \\ 0_1 + \dots + 0_m + 1_m(\frac{1}{m} - 1) + 1_1(\frac{1}{m}) + \dots + 1_{(m-1)}(\frac{1}{m}) \end{bmatrix} \\
&= \begin{bmatrix} m(\frac{1}{m}) \\ \vdots \\ (\frac{1}{m} - 1) + (m-1)(\frac{1}{m}) \\ \vdots \\ (\frac{1}{m} - 1) + (m-1)(\frac{1}{m}) \end{bmatrix}
\end{aligned}$$

Since:

$$m\left(\frac{1}{m}\right) = 1$$

and

$$\begin{aligned}
\left(\frac{1}{m} - 1\right) + (m-1)\frac{1}{m} &= \frac{1}{m} - 1 + \frac{m}{m} - \frac{1}{m} \\
&= -1 + 1 \\
&= 0.
\end{aligned}$$

Then:

$$Q_{2m}\vec{x} = \begin{bmatrix} 1_1 \\ \vdots \\ 1_m \\ 0_1 \\ \vdots \\ 0_m \end{bmatrix}$$

$$= \vec{j}_{2m} - \vec{x}.$$

□

Now that we have shown the necessary lemmas, we will show that adjacency cospectral graphs always result from a GM switch. Recall that similar matrices have the same multiset of eigenvalues so we can prove that two graphs are cospectral by constructing a similarity transformation matrix. We remind the reader of the theorem here.

Theorem 5.4. (Godsil, McKay [5]) *Let G be a graph and let $\pi = (C, D)$ be a partition of $V(G)$ which satisfies conditions*

- (a) $|C| = n$ must be even.
- (b) The induced subgraph formed by the vertex set C must be a regular graph.
- (c) For all $v \in D$, v must be adjacent to either $0, n/2$, or n vertices in C .

Then G and G^π are cospectral graphs.

Proof. Let A be the adjacency matrix of G and let B be the adjacency matrix of G^π where A is labeled in an order that follows $\pi = (C, D)$. Then:

$$A = \begin{bmatrix} A(C) & L \\ L^T & A(D) \end{bmatrix}$$

where L represents the submatrix denoting the adjacencies between vertices across sets C and D . Based on the properties of π we see that $A(C)$ has constant row and constant column sums and that each column of L has either $0, n/2$, or n ones.

We will show that $B = QAQ$ where Q is a block diagonal matrix such that:

$$Q = \begin{bmatrix} Q_n & 0 \\ 0 & I_{|D|} \end{bmatrix}.$$

Next we can examine QQ . We show that $Q = Q^{-1}$ and thus Q is invertible. By the properties of block matrix multiplication:

$$\begin{aligned} QQ &= \begin{bmatrix} Q_n & 0 \\ 0 & I_{|D|} \end{bmatrix} \begin{bmatrix} Q_n & 0 \\ 0 & I_{|D|} \end{bmatrix} \\ &= \begin{bmatrix} (Q_n)^2 & 0 \\ 0 & (I_{|D|})^2 \end{bmatrix} \end{aligned}$$

Then by Lemma 5.1

$$\begin{aligned} QQ &= \begin{bmatrix} I_n & 0 \\ 0 & I_{|D|} \end{bmatrix} \\ &= I. \end{aligned}$$

Lemma 5.2 shows that the adjacencies within the subgraphs of G induced by each of C or D will be preserved by conjugation with Q . Lemma 5.3 shows the change of adjacency between neighbors across vertex sets C and D is immediate on vectors with equal numbers of 0's and 1's when using Q as a transformation matrix.

Q_n is a symmetric matrix and we can see from the block diagonal matrix Q that $Q = Q^T$. That is, Q is also a symmetric matrix. Since $QQ = I$ we can see that $Q = Q^{-1}$. Therefore $Q^{-1} = Q^T$. Then by Lemmas 5.2 and 5.3 the matrix

$$\begin{aligned}
QAQ^{-1} &= \begin{bmatrix} Q_n & 0 \\ 0 & I_{|D|} \end{bmatrix} \begin{bmatrix} A(C) & L \\ L^T & A(D) \end{bmatrix} \begin{bmatrix} Q_n & 0 \\ 0 & I_{|D|} \end{bmatrix} \\
&= \begin{bmatrix} Q_n A(C) & Q_n L \\ I_{|D|} L^T & I_{|D|} A(D) \end{bmatrix} \begin{bmatrix} Q_n & 0 \\ 0 & I_{|D|} \end{bmatrix} \\
&= \begin{bmatrix} Q_n A(C) Q_n & Q_n L I_{|D|} \\ I_{|D|} L^T Q_n & I_{|D|} A(D) I_{|D|} \end{bmatrix} \\
&= \begin{bmatrix} A(C) & L' \\ L^{T'} & A(D) \end{bmatrix}
\end{aligned}$$

where for every column in L that is a zero vector, $Q_n \vec{0} = \vec{0}$. For every column in L such that every entry is 1, $Q_n \vec{1} = \vec{1}$, and by Lemma 5.3 for every column in L with exactly half of its entries equal to 0 and the other half equal to 1, left multiplication by Q_n switches the 1's to 0's and the 0's to 1's. Since we haven't changed the subgraphs induced by C or D , the only difference between A and this matrix are blocks L' and $L^{T'}$.

Observe that L' and $L^{T'}$ represent the vertex adjacencies across sets C and D , and we have only changed entries in columns with half the entries equal to zero. These columns correspond to vertices $v \in D$ that are adjacent to $n/2$ vertices in C . Switching 0's to 1's in these columns turns nonedges across sets into edges, and switching 1's to 0's turns edges into nonedges. Since we do not alter the columns in L' corresponding to vertices $v \in D$ adjacent to all or no vertices in C , we have described a GM switch, and we conclude that $QAQ^{-1} = B$. Thus we have preserved the eigenvalues of A such that A and B will be similar matrices. Since A and B are adjacency matrices of G and G^π respectively, G and G^π are cospectral graphs. $\square$

5.4 Seidel equivalence classes

We have just observed the relationship between Seidel switching and adjacency cospectral graphs. There are a finite number of graphs we can switch to and thus Seidel switching partitions graphs of n vertices into equivalence classes. Consider all possible switching sets for the graph $K_{2,2}$ shown in Figure 14. When we perform a Seidel switch about each of these sets we find that we have constructed a switching class.

For some Seidel classes, $\mathcal{C}$ we can build a new equivalence class $\bar{\mathcal{C}}$ by taking the complement of each graph in $\mathcal{C}$. Here we will refer to $\mathcal{C}$ and $\bar{\mathcal{C}}$ as *complement classes*. Superimposing complement classes gives the complete graph in every case. That is, for all graphs G in a class $\mathcal{C}$, there exists $\bar{G}$ in $\bar{\mathcal{C}}$ such that $E(G) \cup E(\bar{G}) = E(K_n)$. Seidel switching preserves the Seidel characteristic polynomial and spectrum.

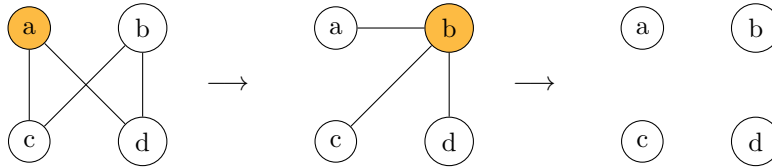


Figure 14: Switching class obtained by switching from $K_{2,2}$ to $K_{1,3}$ with switching set $\{a\}$ and then by switching from $K_{1,3}$ to $K_{0,n}$ with switching set $\{b\}$.

Lemma 5.5. *If $(\lambda, \vec{x})$ is an eigenpair of the Seidel matrix S for a given graph G , then $(-\lambda, \vec{x})$ is an eigenpair of S^c .*

Proof. Let G be a graph. Let S and S^c be the Seidel matrices of G and $\bar{G}$ respectively. Let $(\lambda, \vec{x})$ be an eigenpair of S . Edges in G become non-edges in $\bar{G}$ and the converse is true by definition of a complement. In terms of the Seidel matrix, since edges are denoted by -1 and nonedges by 1 , taking the complement turns every -1 to a 1 and every 1 to a -1 . That is, $S^c = -S$.

By definition of an eigenpair, $S\vec{x} = \lambda\vec{x}$. Then $S^c\vec{x} = -S\vec{x} = -\lambda\vec{x}$ so $(-\lambda, \vec{x})$ is an eigenpair of S^c . $\square$

In the first two sets in Figure 15 are the complete bipartite class and its complement class of graphs. Highlighted in the same color are graphs and their complements. Consider superimposing the graphs of the same color across each class, that is the graph $K_{i,n-i}$ in $\mathcal{C}$ with graph $K_i + K_{n-i}$ in $\bar{\mathcal{C}}$ for $i = 0, \dots, n$. In every case this gives the complete graph. Thus we see that these are complement classes. We will prove in Section 6 that the complete bipartite class of graphs and its complement class of graphs appear in $\mathcal{G}_n$ for all n .

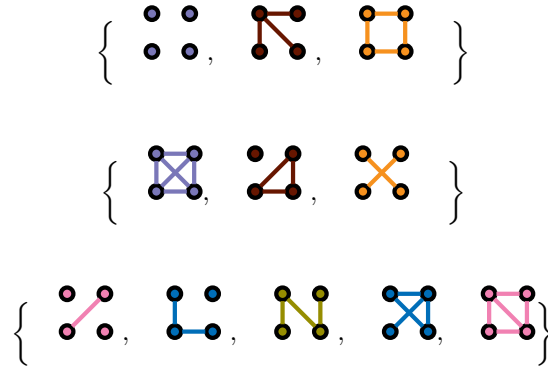


Figure 15: $\mathcal{G}_4$ Seidel classes

A Seidel *switching self-complementary* (SSC) class of graphs is an equivalence class in which any graph in the class is self complementary or has its complement contained within the same class of graphs. An example is shown in the final set of Figure 15 where graphs of the same color are complements. For graphs on n vertices such that $n \equiv 3 \pmod{4}$ there exists no SSC class of graphs [10].

We have now examined properties of specific graphs as well as various graph operations of significance. Using this background we will proceed to present original results on Seidel switching classes and their associated properties.

6 New results on the Seidel matrix

Motivated by strong results on the adjacency matrix and the study of adjacency cospectral graphs, we give results that hold for all graphs before using the Seidel matrix and the associated operation of Seidel switching to show results specific to complete bipartite graphs and their complements.

We proceed by proving results that hold for all Seidel characteristic polynomials. We show a generalized form for the first three coefficients of the Seidel characteristic polynomial for all graphs.

Theorem 6.1. *The coefficients of the terms of the Seidel characteristic polynomial λ^n and λ^{n-1} have magnitudes of 1 and 0, respectively.*

Proof. Let G be a graph with vertex set $[n]$. Let S denote the Seidel matrix of G . By the permutation definition of the determinant

$$\phi(S) = \det(S - \lambda I) = \sum_{\sigma \in S_n} \text{sgn}(\sigma) (s_{1,\sigma(1)} \cdots s_{n,\sigma(n)}) \quad (1)$$

where $s_{i,\sigma(i)}$ for $i = 1, \dots, n$ is an entry of $S - \lambda I$.

For the $n \times n$ matrix $S - \lambda I$ there are n entries on the main diagonal. Due to the structure of the Seidel matrix, for all $i = 1, \dots, n$, $s_{ii} = -\lambda$, and no other λ entry appears in $S - \lambda I$. The matrix entries s_{ii} correspond to fixed elements within a permutation since this indicates element i maps to i . Then every factor s_{ii} that contributes to a term of degree c must be the sum of all permutations with the same number c of fixed elements. To find the magnitude of the coefficient of the degree n term, we first find the number of terms with degree n in equation (1). We consider the number of possible permutations on $[n]$ with n fixed elements, and observe that the only way to obtain a term of degree n comes from the identity permutation since all n values must be fixed. Thus the term λ^n has magnitude 1 in our characteristic polynomial.

To find the number of terms in (1) with degree $n - 1$ we consider the number of possible permutations on $[n]$ with $n - 1$ fixed elements. If we fix $n - 1$ elements, the single remaining element has no possible spot to move to and thus must also be fixed. Therefore, any permutation fixing $n - 1$ elements fixes n elements. Since there are no permutations that fix exactly $n - 1$ elements, there are 0 terms of degree $n - 1$ in the sum given in (1).

Therefore, 1 and 0 are the magnitudes of the coefficients of λ^n and λ^{n-1} of every Seidel characteristic polynomial. □

Theorem 6.2. *The coefficient of the λ^{n-2} term of the Seidel characteristic polynomial has magnitude $\binom{n}{2}$.*

Proof. Following the same logic as in the proof of Theorem 6.1 we examine permutations of $[n]$ that fix $n - 2$ elements.

Observe that there are $\binom{n}{2}$ permutations with $n - 2$ fixed elements. Thus there are $n - 2$ entries of $(-\lambda)$ from the main diagonal of $(S - \lambda I)$ resulting in terms of degree λ^{n-2} . Let $\sigma = (ij)$ for $i \neq j$ be an arbitrary permutation with $n - 2$ fixed elements. Since only two elements move, they must exchange positions. So we claim that each term in (1) with degree $n - 2$ is $\pm \lambda^{n-2}$.

Since the Seidel matrix is symmetric, $S - \lambda I$ is also symmetric. Thus the value of the entries (s_{ij}) and (s_{ji}) are equal. Since the only nondiagonal entries in $S - \lambda I$ are 1 and -1 , $(s_{ij})(s_{ji}) = 1$ for every $i \neq j$. Permutations of $[n]$ with $n - 2$ fixed elements are transpositions and since each of these terms has the same number of fixed elements in our permutation, they have the same $\text{sgn}(\sigma)$.

$$\begin{aligned} \text{sgn}(\sigma)(s_{11} \cdots s_{ij} \cdots s_{ji} \cdots s_{nn}) &= (-1)(-\lambda)(-\lambda) \cdots s_{ij} \cdots s_{ji} \cdots (-\lambda)(-\lambda) \\ &= (-1)(-\lambda)^{n-2}(s_{ij})(s_{ji}) \\ &= (-1)(-1)^{n-2}(\lambda)^{n-2}(s_{ij})(s_{ji}) \\ &= (-1)^{n-1}(\lambda)^{n-2}(s_{ij})(s_{ji}) \\ &= (-1)^{n-1}(\lambda)^{n-2}. \end{aligned}$$

Summing over all pairs of i, j we have $\sum_{i \neq j} (-1)^{n-1} \lambda^{n-2} = \binom{n}{2} (-1)^{n-1} \lambda^{n-2}$. Thus the coefficient of the λ^{n-2} term will have a magnitude of $\binom{n}{2}$. □

We proceed with our examination of complete bipartite graphs with respect to the operation of Seidel switching.

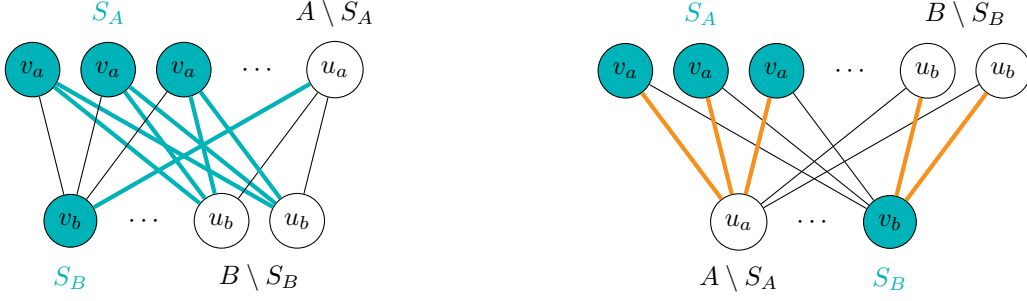


Figure 16: Seidel switch of a complete bipartite graph

Theorem 6.3. *For all n , the set of graphs in the form $K_{a,n-a}$ for $a = 0, \dots, \lfloor \frac{n}{2} \rfloor$ is a Seidel equivalence class.*

Proof. Let C be the set of graphs of the form $K_{a,n-a}$ and let $Q_{K_{a,n-a}}$ be the Seidel equivalence class of $K_{a,n-a}$. Let A and B be vertex partitions for $K_{a,n-a}$ such that $|A| = a$ and $|B| = n - a$. Let S be a subset of $V(K_{a,n-a})$ and since $K_{a,n-a}$ is bipartite we have a partition of S such that $S = S_A \cup S_B$.

Let $v_a \in S_A$, $v_b \in S_B$, $u_a \in A \setminus S_A$, and $u_b \in B \setminus S_B$, as labeled in Figure 16. Note that either S_A or S_B can be empty. In our original graph $K_{a,n-a}$ edges $\{v_a, u_b\}$, $\{v_a, v_b\}$, $\{u_a, v_b\}$, and $\{u_a, u_b\}$ are all present.

We will first show that $Q_{K_{a,n-a}} \subseteq C$. That is, we seek to show that for any switching set S , $(K_{a,n-a})^S$ is a complete bipartite graph.

To perform a Seidel switch $(K_{a,n-a})^S$ we remove all edges of the forms $\{u_a, v_b\}$ and $\{v_a, u_b\}$ then add all edges in the forms $\{v_a, u_a\}$ and $\{v_b, u_b\}$. Thus we have switched from $K_{a,n-a}$ to a graph $K_{a-|S_A|+|S_B|, n-a-|S_B|+|S_A|}$ with edges $\{v_a, v_b\}$, $\{u_a, u_b\}$, $\{v_a, u_a\}$, and $\{v_b, u_b\}$. Since this gives all edges which cross partite sets and none within a partite set, $K_{a-|S_A|+|S_B|, n-a-|S_B|+|S_A|}$ is indeed a complete bipartite graph with partite sets $A \setminus S_A \cup B$ and $B \setminus S_B \cup A$. Therefore $Q_{K_{a,n-a}} \subseteq C$.

Next we seek to show that $C \subseteq Q_{K_{a,n-a}}$. Let $K_{a,n-a}$ be fixed. Let t be arbitrary where $t = 0, \dots, \lfloor \frac{n}{2} \rfloor$. Consider a Seidel switch on the graph $K_{t,n-t}$. We will demonstrate that $K_{t,n-t}$ is in $Q_{K_{a,n-a}}$. That is, we will show that there exists a set S such that $(K_{a,n-a})^S \cong K_{t,n-t}$.

Case 1: Suppose $t = a$. Then we have in effect performed the Seidel switch with $S = \emptyset$ given by $(K_{a,n-a})^S = K_{t,n-t}$.

Case 2: Suppose $t < a$. Then we choose a switching set of any $a - t$ vertices in A . When we perform a Seidel switch with this set our resulting graph is $K_{a-(a-t)+0, n-a-0+(a-t)}$ which reduces to $K_{t,n-t}$.

Case 3: Suppose $t > a$. Then we choose a switching set of any $t - a$ vertices in B . When we perform a Seidel switch our resulting graph is $K_{a-0+(t-a), n-a+0-(t-a)}$ which reduces to $K_{t,n-t}$.

In all cases a Seidel switch $(K_{a,n-a})^S$ gives the graph $K_{t,n-t}$ so we can observe that $(K_{a,n-a})^S \cong K_{t,n-t}$. Thus $C \subseteq Q_{K_{a,n-a}}$.

$Q_{K_{a,n-a}} \subseteq C$ and $C \subseteq Q_{K_{a,n-a}}$, therefore $Q_{K_{a,n-a}} = C$. Thus the set of complete bipartite graphs on n vertices forms a Seidel equivalence class.

□

Next we examine the structure of the Seidel characteristic polynomial for the complete bipartite equivalence class and its complement alongside their spectra. We begin by showing the form of the Seidel spectra for complete bipartite graphs.

Theorem 6.4. *The Seidel spectrum of a complete bipartite graph $K_{a,b}$ is always of the form $\{-1^{(n)}, (n-1)^{(1)}\}$.*

Proof. Consider a complete bipartite graph $K_{a,b}$ with partite sets A, B on n vertices, as shown in Figure 17.

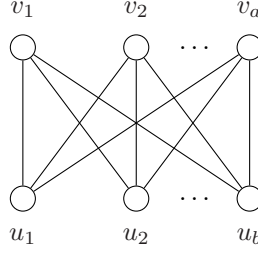


Figure 17: The complete bipartite graph $K_{a,b}$.

If we examine the matrix $(S - \lambda I)$ of this graph, ordered by the partite sets, we have a block matrix:

$$(S(K_{a,b}) - \lambda I) = \begin{bmatrix} -\lambda & 1 & \cdots & 1 & -1 & -1 & \cdots & -1 \\ 1 & -\lambda & \ddots & \vdots & -1 & -1 & \cdots & -1 \\ \vdots & \ddots & \ddots & 1 & \vdots & \ddots & \ddots & \vdots \\ 1 & \cdots & 1 & -\lambda & -1 & \cdots & -1 & -1 \\ -1 & -1 & \cdots & -1 & -\lambda & 1 & \cdots & 1 \\ -1 & -1 & \ddots & \vdots & 1 & -\lambda & \cdots & 1 \\ \vdots & \ddots & \ddots & -1 & \vdots & \ddots & \ddots & \vdots \\ -1 & \cdots & -1 & -1 & 1 & \cdots & 1 & -\lambda \end{bmatrix}$$

$$= \begin{bmatrix} A & B \\ C & D \end{bmatrix}.$$

Both the A and D blocks have the form below:

$$\begin{bmatrix} -\lambda & 1 & \cdots & 1 \\ 1 & -\lambda & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \cdots & 1 & -\lambda \end{bmatrix}$$

where A has dimensions $a \times a$ and D has dimensions $b \times b$. Within blocks A and D we have that all off diagonal entries are equal to 1 since the vertices within a partite set of a complete bipartite graph are non-adjacent. Blocks B and C contain all -1 entries, since these blocks represent pairs of vertices across partite sets. That is, $-J$ blocks give our edges. While both blocks B and C are the matrix $(-J)$ they have differing dimensions. Note that $C = B^T$, which becomes clear when we write out the dimensions of each block based on the partitions given by complete bipartite graphs in their more generalized forms. That is:

$$(S(K_{a,b}) - \lambda I) = \begin{bmatrix} J + I(-1 - \lambda)_{a \times a} & -J_{a \times b} \\ -J_{b \times a} & J + I(-1 - \lambda)_{b \times b} \end{bmatrix}.$$

Since we seek to calculate the determinant of a block matrix, we will use the Schur complement of D , along with Theorem 2.2 to calculate $\det(D)\det(A - BD^{-1}C)$. We begin by calculating $\det(D)$, which has dimensions $b \times b$, using row operations. So we have:

$$\begin{aligned}
\det(D) &= \det \begin{bmatrix} -\lambda & 1 & \cdots & 1 \\ 1 & -\lambda & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \cdots & 1 & -\lambda \end{bmatrix} \\
&= \det \begin{bmatrix} (-\lambda + b - 1) & (-\lambda + b - 1) & \cdots & (-\lambda + b - 1) \\ & 1 & & -\lambda & \ddots & \vdots \\ & \vdots & & \ddots & \ddots & 1 \\ & 1 & & \cdots & 1 & -\lambda \end{bmatrix} \\
&= (-\lambda + b - 1) \det \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & -\lambda & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \cdots & 1 & -\lambda \end{bmatrix} \\
&= (-\lambda + b - 1) \det \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 0 & -\lambda - 1 & 0 & \cdots & 0 \\ \vdots & \ddots & -\lambda - 1 & \ddots & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ 0 & \cdots & \cdots & 0 & -\lambda - 1 \end{bmatrix} \\
&= (-\lambda + b - 1)(1)(-\lambda - 1) \cdots (-\lambda - 1) \\
&= (-\lambda + b - 1)(-\lambda - 1)^{b-1}.
\end{aligned}$$

Next we consider $\det(A - BD^{-1}C)$. We will begin by calculating D^{-1} . Note that we can rewrite D such that $D = P + Q$. Recall that $D = J - \lambda I - I$, so we can let $P = I(-\lambda - 1)$ and $Q = J$. Then the hypotheses of Theorem 2.1 are satisfied and:

$$\begin{aligned}
(P + Q)^{-1} &= P^{-1} - \frac{1}{1 + \text{tr}(QP^{-1})} P^{-1} Q P^{-1} \\
&= (I(-\lambda - 1))^{-1} - \frac{1}{1 + \text{tr}(J(I(-\lambda - 1))^{-1})} (I(-\lambda - 1))^{-1} J (I(-\lambda - 1))^{-1} \\
&= I \left(\frac{1}{-\lambda - 1} \right) - \left(\frac{1}{1 + \left(\frac{b}{-\lambda - 1} \right)} I \left(\frac{1}{-\lambda - 1} \right) J I \left(\frac{1}{-\lambda - 1} \right) \right)
\end{aligned}$$

$$\begin{aligned}
&= I \left(\frac{1}{-\lambda-1} \right) - \left(\frac{1}{\frac{-\lambda-1+b}{-\lambda-1}} \left(\frac{1}{-\lambda-1} \right) J \left(\frac{1}{-\lambda-1} \right) \right) \\
&= \left(\frac{1}{-\lambda-1} \right) \left(I - \frac{1}{\frac{-\lambda-1+b}{-\lambda-1}} \left(\frac{1}{-\lambda-1} \right) J \right) \\
&= \left(\frac{1}{-\lambda-1} \right) \left(I - \frac{J}{-\lambda-1+b} \right) \\
&= \left(\frac{1}{-\lambda-1} \right) \left(I - \begin{bmatrix} \frac{1}{-\lambda-1+b} & \cdots & \frac{1}{-\lambda-1+b} \\ \vdots & \ddots & \vdots \\ \frac{1}{-\lambda-1+b} & \cdots & \frac{1}{-\lambda-1+b} \end{bmatrix} \right) \\
&= \left(\frac{1}{-\lambda-1} \right) \begin{bmatrix} 1 - \frac{1}{-\lambda-1+b} & \frac{-1}{-\lambda-1+b} & \cdots & \frac{-1}{-\lambda-1+b} \\ \frac{-1}{-\lambda-1+b} & 1 - \frac{1}{-\lambda-1+b} & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{-1}{-\lambda-1+b} \\ \frac{-1}{-\lambda-1+b} & \cdots & \frac{-1}{-\lambda-1+b} & 1 - \frac{1}{-\lambda-1+b} \end{bmatrix} \\
&= \left(\frac{1}{-\lambda-1} \right) \begin{bmatrix} \left(\frac{-\lambda-2+b}{-\lambda-1+b} \right) & \frac{-1}{-\lambda-1+b} & \cdots & \frac{-1}{-\lambda-1+b} \\ \frac{-1}{-\lambda-1+b} & \left(\frac{-\lambda-2+b}{-\lambda-1+b} \right) & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{-1}{-\lambda-1+b} \\ \frac{-1}{-\lambda-1+b} & \cdots & \frac{-1}{-\lambda-1+b} & \left(\frac{-\lambda-2+b}{-\lambda-1+b} \right) \end{bmatrix} \\
&= \left(\frac{1}{-\lambda-1} \right) \left(\frac{1}{-\lambda-1+b} \right) \begin{bmatrix} -\lambda-2+b & -1 & \cdots & -1 \\ -1 & -\lambda-2+b & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \cdots & -1 & -\lambda-2+b \end{bmatrix} \\
&= \frac{1}{(-\lambda-1)(-\lambda-1+b)} \begin{bmatrix} -\lambda-2+b & -1 & \cdots & -1 \\ -1 & -\lambda-2+b & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \cdots & -1 & -\lambda-2+b \end{bmatrix}.
\end{aligned}$$

Now we will compute $BD^{-1}C$. Note that B has dimensions $a \times b$, D^{-1} has dimensions $b \times b$, and C has

dimensions $b \times a$. Thus our matrices are conforming and the product will be a matrix with dimensions $a \times a$.

$$\begin{aligned}
BD^{-1}C &= \begin{bmatrix} -1 & \cdots & -1 \\ \vdots & \ddots & \vdots \\ -1 & \cdots & -1 \end{bmatrix} \frac{1}{(-\lambda-1)(-\lambda-1+b)} \begin{bmatrix} -\lambda-2+b & -1 & \cdots & -1 \\ -1 & -\lambda-2+b & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \cdots & -1 & -\lambda-2+b \end{bmatrix} \begin{bmatrix} -1 & \cdots & -1 \\ \vdots & \ddots & \vdots \\ -1 & \cdots & -1 \end{bmatrix} \\
&= \frac{1}{(-\lambda-1)(-\lambda-1+b)} \begin{bmatrix} -1 & \cdots & -1 \\ \vdots & \ddots & \vdots \\ -1 & \cdots & -1 \end{bmatrix} \begin{bmatrix} -\lambda-2+b & -1 & \cdots & -1 \\ -1 & -\lambda-2+b & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \cdots & -1 & -\lambda-2+b \end{bmatrix} \begin{bmatrix} -1 & \cdots & -1 \\ \vdots & \ddots & \vdots \\ -1 & \cdots & -1 \end{bmatrix} \\
&= \frac{1}{(-\lambda-1)(-\lambda-1+b)} \begin{bmatrix} \lambda+1 & \cdots & \lambda+1 \\ \vdots & \ddots & \vdots \\ \lambda+1 & \cdots & \lambda+1 \end{bmatrix} \begin{bmatrix} -1 & \cdots & -1 \\ \vdots & \ddots & \vdots \\ -1 & \cdots & -1 \end{bmatrix} \\
&= \frac{1}{(-\lambda-1)(-\lambda-1+b)} \begin{bmatrix} b(-\lambda-1) & \cdots & b(-\lambda-1) \\ \vdots & \ddots & \vdots \\ b(-\lambda-1) & \cdots & b(-\lambda-1) \end{bmatrix} \\
&= \begin{bmatrix} \frac{b}{-\lambda-1+b} & \cdots & \frac{b}{-\lambda-1+b} \\ \vdots & \ddots & \vdots \\ \frac{b}{-\lambda-1+b} & \cdots & \frac{b}{-\lambda-1+b} \end{bmatrix}.
\end{aligned}$$

Finally, we calculate $A - BD^{-1}C$. Let m_{ij} denote the entries of $A - BD^{-1}C$. These entries are given by

$$m_{ij} = \begin{cases} \frac{\lambda^2 - b\lambda + \lambda - b}{-\lambda - 1 + b} & \text{if } i = j \\ \frac{-\lambda - 1}{-\lambda - 1 + b} & \text{if } i \neq j. \end{cases}$$

Then, the determinant of the Schur complement of D gives:

$$\det(A - BD^{-1}C) = \det \begin{bmatrix} \frac{\lambda^2 - b\lambda + \lambda - b}{-\lambda - 1 + b} & \frac{-\lambda - 1}{-\lambda - 1 + b} & \cdots & \frac{-\lambda - 1}{-\lambda - 1 + b} \\ \frac{-\lambda - 1}{-\lambda - 1 + b} & \frac{\lambda^2 - b\lambda + \lambda - b}{-\lambda - 1 + b} & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{-\lambda - 1}{-\lambda - 1 + b} \\ \frac{-\lambda - 1}{-\lambda - 1 + b} & \cdots & \frac{-\lambda - 1}{-\lambda - 1 + b} & \frac{\lambda^2 - b\lambda + \lambda - b}{-\lambda - 1 + b} \end{bmatrix}.$$

We now add all rows to row 1 then scale row 1 by $\frac{1}{c}$ for $c = \left(\frac{\lambda^2 - b\lambda + \lambda - b}{-\lambda - 1 + b} + (a-1) \frac{-\lambda - 1}{-\lambda - 1 + b} \right)$. Then:

$$\det(A - BD^{-1}C) = c \cdot \det \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ \frac{-\lambda - 1}{-\lambda - 1 + b} & \frac{\lambda^2 - b\lambda + \lambda - b}{-\lambda - 1 + b} & \frac{-\lambda - 1}{-\lambda - 1 + b} & \cdots & \frac{-\lambda - 1}{-\lambda - 1 + b} \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & \frac{-\lambda - 1}{-\lambda - 1 + b} \\ \frac{-\lambda - 1}{-\lambda - 1 + b} & \cdots & \cdots & \frac{-\lambda - 1}{-\lambda - 1 + b} & \frac{\lambda^2 - b\lambda + \lambda - b}{-\lambda - 1 + b} \end{bmatrix}$$

$$\begin{aligned}
&= c \cdot \det \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 0 & \frac{\lambda^2 - b\lambda + \lambda - b}{-\lambda - 1 + b} - \frac{-\lambda - 1}{-\lambda - 1 + b} & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & \frac{\lambda^2 - b\lambda + \lambda - b}{-\lambda - 1 + b} - \frac{-\lambda - 1}{-\lambda - 1 + b} \end{bmatrix} \\
&= c \cdot \left(\frac{\lambda^2 - b\lambda + \lambda - b}{-\lambda + b - 1} - \frac{-\lambda - 1}{-\lambda + b - 1} \right)^{(a-1)} \\
&= c \cdot \left(\frac{\lambda^2 - b\lambda + 2\lambda - b + 1}{-\lambda + b - 1} \right)^{(a-1)} \\
&= \left(\frac{\lambda^2 - b\lambda + \lambda - b}{-\lambda - 1 + b} + (a-1) \frac{-\lambda - 1}{-\lambda - 1 + b} \right) \cdot \left(\frac{\lambda^2 - b\lambda + 2\lambda - b + 1}{-\lambda + b - 1} \right)^{(a-1)} \\
&= \frac{\lambda^2 - b\lambda + 2\lambda - b - a\lambda - a + 1}{-\lambda + b - 1} \cdot \left(\frac{\lambda^2 - b\lambda + 2\lambda - b + 1}{-\lambda + b - 1} \right)^{(a-1)}.
\end{aligned}$$

Now we consider the full formula for the determinant of a block matrix. For these calculations, recall that $a + b = n$, the number of vertices in the complete bipartite graph $K_{a,b}$. Then:

$$\begin{aligned}
\det(D) \det(A - BD^{-1}C) &= (-\lambda + b - 1)(-\lambda - 1)^{(b-1)} \cdot \frac{\lambda^2 - b\lambda + 2\lambda - b - a\lambda - a + 1}{-\lambda + b - 1} \cdot \left(\frac{\lambda^2 - b\lambda + 2\lambda - b + 1}{-\lambda + b - 1} \right)^{(a-1)} \\
&= (-\lambda - 1)^{(b-1)} \cdot (\lambda^2 - b\lambda + 2\lambda - b - a\lambda - a + 1) \cdot \left(\frac{(-\lambda - 1)(-\lambda + b - 1)}{(-\lambda + b - 1)} \right)^{(a-1)} \\
&= (-\lambda - 1)^{(b-1)} \cdot (\lambda^2 + \lambda(2 - b - a) + 1(1 - b - a)) \cdot (-\lambda - 1)^{(a-1)} \\
&= (-\lambda - 1)^{(a+b-2)} (\lambda^2 + \lambda(2 - (b + a)) + 1(1 - (b + a))) \\
&= (-\lambda - 1)^{(n-2)} (\lambda^2 + \lambda(2 - (n)) + 1(1 - (n))) \\
&= (-\lambda - 1)^{(n-2)} (\lambda^2 + 2\lambda + 1 - n\lambda - n) \\
&= (-\lambda - 1)^{(n-2)} ((-\lambda - 1)^2 + n(-\lambda - 1)) \\
&= (-\lambda - 1)^{(n)} + n(-\lambda - 1)^{(n-1)} \\
&= (-\lambda - 1)^{(n-1)} (-\lambda - 1 + n).
\end{aligned}$$

This gives the Seidel spectrum $\{-1^{(n-1)}, (n-1)^{(1)}\}$. □

Observe that this is also known to be the spectrum of K_n with respect to the adjacency matrix. We now give a form of the Seidel characteristic polynomial for the complete bipartite class.

Corollary 6.4.1. *The Seidel characteristic polynomial of a complete bipartite graph is always given by the form:*

$$\sum_{k=0}^n (-1) \binom{n}{k} (k-1) x^{n-k}.$$

Proof. We will consider the binomial theorem and manipulate it to arrive at the factored form of the complete bipartite Seidel characteristic polynomial. We begin by multiplying the binomial theorem by y^{-1} .

$$\sum_{k=0}^n \binom{n}{k} x^{n-k} y^k = (x+y)^n$$

$$y^{-1} \left(\sum_{k=0}^n \binom{n}{k} x^{n-k} y^k = (x+y)^n \right)$$

$$y^{-1} \left(\sum_{k=0}^n \binom{n}{k} x^{n-k} y^k \right) = y^{-1} (x+y)^n$$

$$\sum_{k=0}^n \binom{n}{k} x^{n-k} y^k y^{-1} = y^{-1} (x+y)^n$$

$$\sum_{k=0}^n \binom{n}{k} x^{n-k} y^{k-1} = y^{-1} (x+y)^n.$$

From here, we calculate the derivative with respect to y . That is,

$$\frac{d}{dy} \left(\sum_{k=0}^n \binom{n}{k} x^{n-k} y^{k-1} = y^{-1} (x+y)^n \right)$$

$$\sum_{k=0}^n \binom{n}{k} x^{n-k} (k-1) y^{k-2} = y^{-1} n (x+y)^{n-1} + (x+y)^n \cdot -y^{-2}$$

Then when we let $y = 1$ and $x = \lambda$. We have,

$$\sum_{k=0}^n \binom{n}{k} \lambda^{n-k} (k-1) 1^{k-2} = 1^{-1} n (\lambda+1)^{n-1} - (\lambda+1)^n \cdot 1^{-2}$$

$$\sum_{k=0}^n \binom{n}{k} \lambda^{n-k} (k-1) = n (\lambda+1)^{n-1} - (\lambda+1)^n.$$

Finally, we multiply the expression by (-1) .

$$-1 \cdot \sum_{k=0}^n \binom{n}{k} \lambda^{n-k} (k-1) = -1 \cdot (n (\lambda+1)^{n-1} - (\lambda+1)^n)$$

$$\sum_{k=0}^n (-1) \binom{n}{k} \lambda^{n-k} (k-1) = -n (\lambda+1)^{n-1} + (\lambda+1)^n$$

$$\sum_{k=0}^n (-1) \binom{n}{k} \lambda^{n-k} (k-1) = (\lambda+1)^n - n(\lambda+1)^{n-1}$$

$$\sum_{k=0}^n (-1) \binom{n}{k} \lambda^{n-k} (k-1) = (\lambda+1)^{n-1} (\lambda+1-n)$$

We have $(\lambda+1)^{n-1}(\lambda+1-n)$ which gives the Seidel characteristic polynomial of a complete bipartite graph as shown in the proof of Theorem 6.4. $\square$

Observe that the complement of a complete bipartite graph is a disjoint union of complete components graph with two complete components. We will show a general form for the Seidel spectrum of this class of graphs. We give a detailed proof of the following corollary, but we note that it is an immediate application of Theorem 6.4 and Lemma 5.5.

Corollary 6.4.2. *The Seidel spectrum of the complement of a complete bipartite graph, $K_a + K_b$ is always of the form $\{1^{(n)}, (1-n)^{(1)}\}$.*

Proof. Consider the matrix $S - \lambda I$ of a graph $K_a + K_b$. We can then rewrite this as a block matrix partitioned by components K_a and K_b such that we have blocks A, B, C, D with dimensions $a \times a, a \times b, b \times a$, and $b \times b$ respectively.

$$(S(K_a + K_b) - \lambda I) = \begin{bmatrix} -\lambda & -1 & \cdots & -1 & 1 & 1 & \cdots & 1 \\ -1 & -\lambda & \ddots & \vdots & 1 & 1 & \cdots & 1 \\ \vdots & \ddots & \ddots & -1 & \vdots & \ddots & \ddots & \vdots \\ -1 & \cdots & -1 & -\lambda & 1 & \cdots & 1 & 1 \\ 1 & 1 & \cdots & 1 & -\lambda & -1 & \cdots & -1 \\ 1 & 1 & \ddots & \vdots & -1 & -\lambda & \cdots & -1 \\ \vdots & \ddots & \ddots & 1 & \vdots & \ddots & \ddots & \vdots \\ 1 & \cdots & 1 & 1 & -1 & \cdots & -1 & -\lambda \end{bmatrix}$$

$$= \begin{bmatrix} A & B \\ C & D \end{bmatrix}.$$

We will proceed using similar methods to those in Theorem 6.4 with the Schur complement of D which has dimensions $b \times b$. Beginning with $\det(D)$ we have:

$$\det(D) = \det \begin{bmatrix} -\lambda & -1 & \cdots & -1 \\ -1 & -\lambda & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \cdots & -\lambda & -\lambda \end{bmatrix}$$

$$= \det \begin{bmatrix} (-\lambda - b + 1) & (-\lambda - b + 1) & \cdots & (-\lambda - b + 1) \\ -1 & -\lambda & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \cdots & -\lambda & -\lambda \end{bmatrix}$$

$$\begin{aligned}
&= (-\lambda - b + 1) \det \begin{bmatrix} 1 & 1 & \cdots & 1 \\ -1 & -\lambda & \ddots & \vdots \\ \vdots & \ddots & \ddots & -1 \\ -1 & \cdots & -\lambda & -\lambda \end{bmatrix} \\
&= (-\lambda - b + 1) \det \begin{bmatrix} 1 & 1 & 1 & \cdots & 1 \\ 0 & -\lambda + 1 & 0 & \cdots & 0 \\ \vdots & \ddots & -\lambda + 1 & \ddots & \vdots \\ \vdots & & \ddots & \ddots & 0 \\ 0 & \cdots & \cdots & 0 & -\lambda + 1 \end{bmatrix} \\
&= (-\lambda - b + 1)(1)(-\lambda + 1) \cdots (-\lambda + 1) \\
&= (-\lambda - b + 1)(-\lambda + 1)^{b-1}.
\end{aligned}$$

From here we calculate D^{-1} and note that $D = (1 - \lambda)I + (-J)$. Let $P = (1 - \lambda)I$ and $Q = -J$. Then by Lemma 2.1

$$\begin{aligned}
(P + Q)^{-1} &= P^{-1} - \frac{1}{1 + \text{tr}(QP^{-1})} P^{-1} Q P^{-1} \\
&= (I(1 - \lambda))^{-1} - \frac{1}{1 + \text{tr}(-J(I(1 - \lambda))^{-1})} (I(1 - \lambda))^{-1} J (I(1 - \lambda))^{-1} \\
&= I \left(\frac{1}{1 - \lambda} \right) - \left(\frac{1}{1 + \left(\frac{-b}{1 - \lambda} \right)} I \left(\frac{1}{1 - \lambda} \right) (-J) I \left(\frac{1}{1 - \lambda} \right) \right) \\
&= I \left(\frac{1}{1 - \lambda} \right) - \left(\frac{1}{\frac{1 - \lambda - b}{1 - \lambda}} \left(\frac{1}{1 - \lambda} \right) (-J) \left(\frac{1}{1 - \lambda} \right) \right) \\
&= \left(\frac{1}{1 - \lambda} \right) \left(I - \frac{1}{\frac{1 - \lambda - b}{1 - \lambda}} \left(\frac{1}{1 - \lambda} \right) (-J) \right) \\
&= \left(\frac{1}{1 - \lambda} \right) \left(I + \frac{J}{1 - \lambda - b} \right) \\
&= \left(\frac{1}{1 - \lambda} \right) \left(I + \begin{bmatrix} \frac{1}{-\lambda + 1 - b} & \cdots & \frac{1}{-\lambda + 1 - b} \\ \vdots & \ddots & \vdots \\ \frac{1}{-\lambda + 1 - b} & \cdots & \frac{1}{-\lambda + 1 - b} \end{bmatrix} \right)
\end{aligned}$$

$$\begin{aligned}
&= \left(\frac{1}{1-\lambda} \right) \begin{bmatrix} 1 + \frac{1}{-\lambda+1-b} & \frac{1}{-\lambda+1-b} & \cdots & \frac{1}{-\lambda+1-b} \\ \frac{1}{-\lambda+1-b} & 1 + \frac{1}{-\lambda+1-b} & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{1}{-\lambda+1-b} \\ \frac{1}{-\lambda+1-b} & \cdots & \frac{1}{-\lambda+1-b} & 1 + \frac{1}{-\lambda+1-b} \end{bmatrix} \\
&= \frac{1}{(1-\lambda)(-\lambda+1-b)} \begin{bmatrix} -\lambda+2-b & 1 & \cdots & 1 \\ 1 & -\lambda+2-b & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \cdots & 1 & -\lambda+2-b \end{bmatrix}.
\end{aligned}$$

Observe that B has dimensions $a \times b$, D^{-1} has dimensions $b \times b$, and C has dimensions $b \times a$. Thus the product $BD^{-1}C$ will be a matrix with dimensions $a \times a$.

$$\begin{aligned}
BD^{-1}C &= \begin{bmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{bmatrix} \frac{1}{(1-\lambda)(-\lambda+1-b)} \begin{bmatrix} -\lambda+2-b & 1 & \cdots & 1 \\ 1 & -\lambda+2-b & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \cdots & 1 & -\lambda+2-b \end{bmatrix} \begin{bmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{bmatrix} \\
&= \frac{1}{(1-\lambda)(-\lambda+1-b)} \begin{bmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{bmatrix} \begin{bmatrix} -\lambda+2-b & 1 & \cdots & 1 \\ 1 & -\lambda+2-b & \ddots & \vdots \\ \vdots & \ddots & \ddots & 1 \\ 1 & \cdots & 1 & -\lambda+2-b \end{bmatrix} \begin{bmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{bmatrix} \\
&= \frac{1}{(1-\lambda)(-\lambda+1-b)} \begin{bmatrix} -\lambda+1 & \cdots & -\lambda+1 \\ \vdots & \ddots & \vdots \\ -\lambda+1 & \cdots & -\lambda+1 \end{bmatrix} \begin{bmatrix} 1 & \cdots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \cdots & 1 \end{bmatrix} \\
&= \frac{1}{(1-\lambda)(-\lambda+1-b)} \begin{bmatrix} b(-\lambda+1) & \cdots & b(-\lambda+1) \\ \vdots & \ddots & \vdots \\ b(-\lambda+1) & \cdots & b(-\lambda+1) \end{bmatrix} \\
&= \begin{bmatrix} \frac{b}{-\lambda+1-b} & \cdots & \frac{b}{-\lambda+1-b} \\ \vdots & \ddots & \vdots \\ \frac{b}{-\lambda+1-b} & \cdots & \frac{b}{-\lambda+1-b} \end{bmatrix}.
\end{aligned}$$

Now we calculate $A - BD^{-1}C$. Let m_{ij} denote the entries of $A - BD^{-1}C$. These entries are given by

$$m_{ij} = \begin{cases} \frac{\lambda^2+b\lambda-\lambda-b}{-\lambda+1-b} & \text{if } i = j \\ \frac{\lambda-1}{-\lambda+1b} & \text{if } i \neq j. \end{cases}$$

Then,

$$\det(A - BD^{-1}C) = \det \begin{bmatrix} \frac{\lambda^2 + b\lambda - \lambda - b}{-\lambda + 1 - b} & \frac{\lambda - 1}{-\lambda + 1 - b} & \cdots & \frac{\lambda - 1}{-\lambda + 1 - b} \\ \frac{\lambda - 1}{-\lambda + 1 - b} & \frac{\lambda^2 + b\lambda - \lambda - b}{-\lambda + 1 - b} & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\lambda - 1}{-\lambda + 1 - b} \\ \frac{\lambda - 1}{-\lambda + 1 - b} & \cdots & \frac{\lambda - 1}{-\lambda + 1 - b} & \frac{\lambda^2 + b\lambda - \lambda - b}{-\lambda + 1 - b} \end{bmatrix}.$$

Let $\left(\frac{\lambda^2 + b\lambda - \lambda - b}{-\lambda + 1 - b} + (a - 1)\frac{\lambda - 1}{-\lambda + 1 - b}\right) = c$. When we add all rows to row 1 we have

$$\begin{aligned} \det(A - BD^{-1}C) &= \det \begin{bmatrix} c & c & \cdots & c \\ \frac{\lambda - 1}{-\lambda + 1 - b} & \frac{\lambda^2 + b\lambda - \lambda - b}{-\lambda + 1 - b} & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\lambda - 1}{-\lambda + 1 - b} \\ \frac{\lambda - 1}{-\lambda + 1 - b} & \cdots & \frac{\lambda - 1}{-\lambda + 1 - b} & \frac{\lambda^2 + b\lambda - \lambda - b}{-\lambda + 1 - b} \end{bmatrix} \\ &= c \cdot \det \begin{bmatrix} 1 & 1 & \cdots & 1 \\ \frac{\lambda - 1}{-\lambda + 1 - b} & \frac{\lambda^2 + b\lambda - \lambda - b}{-\lambda + 1 - b} & \ddots & \vdots \\ \vdots & \ddots & \ddots & \frac{\lambda - 1}{-\lambda + 1 - b} \\ \frac{\lambda - 1}{-\lambda + 1 - b} & \cdots & \frac{\lambda - 1}{-\lambda + 1 - b} & \frac{\lambda^2 + b\lambda - \lambda - b}{-\lambda + 1 - b} \end{bmatrix} \\ &= c \cdot \det \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 0 & \frac{\lambda^2 + b\lambda - \lambda - b}{-\lambda + 1 - b} - \frac{\lambda - 1}{-\lambda + 1 - b} & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \frac{\lambda^2 + b\lambda - \lambda - b}{-\lambda + 1 - b} - \frac{\lambda - 1}{-\lambda + 1 - b} \end{bmatrix} \\ &= c \cdot \left(\frac{\lambda^2 + b\lambda - \lambda - b - \lambda + 1}{-\lambda - b + 1} \right)^{(a-1)} \\ &= c \cdot \left(\frac{\lambda^2 + b\lambda - 2\lambda - b + 1}{-\lambda - b + 1} \right)^{(a-1)} \\ &= \left(\frac{\lambda^2 + b\lambda - \lambda - b + (a - 1)(\lambda - 1)}{-\lambda - b + 1} \right) \cdot \left(\frac{\lambda^2 + b\lambda - 2\lambda - b + 1}{-\lambda - b + 1} \right)^{(a-1)} \\ &= \frac{\lambda^2 + b\lambda - 2\lambda - b + a\lambda - a + 1}{-\lambda - b + 1} \cdot \left(\frac{(-\lambda + 1)(-\lambda - b + 1)}{-\lambda - b + 1} \right)^{(a-1)}. \end{aligned}$$

Now we consider the full formula for the determinant of a block matrix. That is:

$$\begin{aligned} \det(D) \det(A - BD^{-1}C) &= (-\lambda - b + 1)(-\lambda + 1)^{(b-1)} \cdot \frac{\lambda^2 + b\lambda - 2\lambda - b + a\lambda - a + 1}{-\lambda - b + 1} \cdot \left(\frac{(-\lambda + 1)(-\lambda - b + 1)}{-\lambda - b + 1} \right)^{(a-1)} \\ &= (-\lambda + 1)^{(b-1)} \cdot (\lambda^2 + b\lambda - 2\lambda + a\lambda - b - a + 1) \cdot (-\lambda + 1)^{(a-1)} \\ &= (-\lambda + 1)^{(b-1)} \cdot (\lambda^2 + \lambda(b + a - 2) - 1(b + a - 1)) \cdot (-\lambda + 1)^{(a-1)} \end{aligned}$$

$$\begin{aligned}
&= (-\lambda + 1)^{(a+b-2)}(\lambda^2 + \lambda((b+a) - 2) - 1((b+a) - 1)) \\
&= (-\lambda + 1)^{(n-2)}(\lambda^2 + \lambda(n-2) - 1(n-1)) \\
&= (-\lambda + 1)^{(n-2)}(\lambda^2 + n\lambda - 2\lambda - n + 1) \\
&= (-\lambda + 1)^{(n-2)}((\lambda^2 - 2\lambda + 1) + n(\lambda - 1)) \\
&= (-1)^{(n-2)}(\lambda - 1)^{(n-2)}((\lambda - 1)^{(2)} + n(\lambda - 1)) \\
&= (-1)^{(n-2)}((\lambda - 1)^{(n)} + n(\lambda - 1)^{(n-1)}) \\
&= (-1)^{(n-2)}((\lambda - 1)^{(n-1)}(\lambda - 1 + n)).
\end{aligned}$$

This gives the Seidel spectrum $\{1^{(n-1)}, (1-n)^{(1)}\}$. □

We now give a form of the Seidel characteristic polynomial for the complement class of the complete bipartite class which contains graphs made up of two complete components.

Corollary 6.4.3. *The Seidel characteristic polynomial of graphs in the complete bipartite complement class is always given by the form:*

$$\sum_{k=0}^n (-1)^{k+1} \binom{n}{k} (k-1)x^{n-k}.$$

Proof. Consider the binomial theorem:

$$\begin{aligned}
\sum_{k=0}^n \binom{n}{k} x^{n-k} y^k &= (x+y)^n \\
y^{-1} \left(\sum_{k=0}^n \binom{n}{k} x^{n-k} y^k \right) &= (x+y)^n \\
y^{-1} \left(\sum_{k=0}^n \binom{n}{k} x^{n-k} y^k \right) &= y^{-1} (x+y)^n \\
\sum_{k=0}^n \binom{n}{k} x^{n-k} y^k y^{-1} &= y^{-1} (x+y)^n \\
\sum_{k=0}^n \binom{n}{k} x^{n-k} y^{k-1} &= y^{-1} (x+y)^n \\
\frac{d}{dy} \left(\sum_{k=0}^n \binom{n}{k} x^{n-k} y^{k-1} \right) &= y^{-1} (x+y)^n
\end{aligned}$$

$$\sum_{k=0}^n \binom{n}{k} x^{n-k} (k-1) y^{k-2} = y^{-1} n (x+y)^{n-1} + (x+y)^n \cdot -y^{-2}$$

$$y^3 \left(\sum_{k=0}^n \binom{n}{k} x^{n-k} (k-1) y^{k-2} = y^{-1} n (x+y)^{n-1} + (x+y)^n \cdot -y^{-2} \right)$$

$$\sum_{k=0}^n \binom{n}{k} x^{n-k} (k-1) y^{k+1} = y^2 n (x+y)^{n-1} + (x+y)^n \cdot -y.$$

Let $y = -1$ and $x = \lambda$. Then,

$$\sum_{k=0}^n \binom{n}{k} \lambda^{n-k} (k-1) (-1)^{k+1} = (-1)^2 n (\lambda-1)^{n-1} + (\lambda-1)^n \cdot (-1)^2$$

$$\sum_{k=0}^n \binom{n}{k} \lambda^{n-k} (k-1) (-1)^{k+1} = n (\lambda-1)^{n-1} + (\lambda-1)^n$$

$$\sum_{k=0}^n \binom{n}{k} \lambda^{n-k} (k-1) (-1)^{k+1} = (\lambda-1)^{n-1} (\lambda-1+n).$$

So we have $(\lambda-1)^{n-1} (\lambda-1+n)$, which gives the Seidel characteristic polynomial of the complement of a complete bipartite graph, as shown in Corollary 6.4.3. □

Here we have given a form for the magnitudes of the first three coefficients of every Seidel characteristic polynomial. Next we have shown that there exists a complete bipartite switching class for Seidel classes on any n vertices in Theorem 6.3. We provided a characterization of the Seidel spectrum for complete bipartite graphs and their graph complements in Theorem 6.4 and Corollary 6.4.2. We concluded our results by giving a form for the characteristic polynomials of these classes in Corollaries 6.4.1 and 6.4.3.

7 Conclusion

In this paper we have examined cospectral graphs with respect to the adjacency and Seidel matrices. We have explained in detail the original proof of the GM-switch. We have also provided a general form of the Seidel characteristic polynomial, $\phi(S(G))$ for all complete bipartite graphs on n vertices and their complements.

For future research directions we would be interested in searching the literature on self containing classes for open questions, where self containing Seidel classes are equivalence classes which contain their graph complements. We hope that this could provide insights into what structural similarities graphs in classes without a complement class possess. Similarly, it would be interesting to study graphs in particular complement classes in the hopes of determining a clear structural property shared by all graphs within a class, as was the case for the complete bipartite class and its complement class. We have observed a pattern for golden graphs, graphs that possess at least one eigenvalue equal to the golden ratio. Based on computational evidence, we conjecture that for any $n > 5$ in which golden graphs exist, there is a Seidel class containing only golden graphs.

Other research directions could include an examination of the distribution of adjacency cospectral graphs across Seidel equivalence classes in order to better determine graph properties that give non-DS graphs.

Because of the relationship between GM switching and Seidel switching, we conjecture that such adjacency cospectral graphs cannot be obtained through a GM switch.

Finally, we could look into determining a generalizable form for the fourth coefficient of $\phi(S(G))$. The third coefficient gives the total number of possible edges of G for any graph on n vertices. Therefore it is our hope that the fourth coefficient of $\phi(S(G))$ has some discernible relationship to graphs on n vertices and in particular the switching class containing G .

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